

CIVL6410 Transport Network Modelling

# Week 1: Introduction

A/Prof Jiwon Kim  
Semester 1, 2026

# Course Administrative Details

# Course Timetable



## CIVL6410 Transport Network Modelling Timetable: Semester 1, 2026

|                                                                       | Week 1<br>23-Feb-26 | Week 2<br>02-Mar-26 | Week 3<br>09-Mar-26 | Week 4<br>16-Mar-26 | Week 5<br>23-Mar-26 | Week 6<br>30-Mar-26 | Sem.Break<br>06-Apr-26 | Week 7<br>13-Apr-26 | Week 8<br>20-Apr-26 | Week 9<br>27-Apr-26 | Week 10<br>04-May-26 | Week 11<br>11-May-26 | Week 12<br>18-May-26 | Week 13<br>25-May-26 |
|-----------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|------------------------|---------------------|---------------------|---------------------|----------------------|----------------------|----------------------|----------------------|
| Activities                                                            | M T W T F           | M T W T F           | M T W T F           | M T W T F           | M T W T F           | M T W T F           | M T W T F              | M T W T F           | M T W T F           | M T W T F           | M T W T F            | M T W T F            | M T W T F            | M T W T F            |
| <b>Lectures</b><br>Tue 10am-12pm (47A-352)                            |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |
| <b>Tutorials</b><br>Wed 10-11am (69-304)                              |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |
| <b>In-class Quizzes</b><br>During Tutorials                           |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |
| <b>Assignments</b><br>Two assignments,<br>each with a 2-week deadline |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |

public holiday or semester break

- ▶ **Lectures:** 2h in-person session
  - Concepts, methodologies, and application examples
  - Lecture recordings will be posted on the Blackboard.
- ▶ **Tutorials:** 1h in-person session
  - **Python exercises** and **in-class quizzes**
  - Tutorials will **not** be recorded.

|             | Mon | Tue     | Wed                | Thu | Fri |
|-------------|-----|---------|--------------------|-----|-----|
| 8am - 9am   |     |         |                    |     |     |
| 9am - 10am  |     |         |                    |     |     |
| 10am - 11am |     | Lecture | Tutorial<br>69-304 |     |     |
| 11am - 12pm |     | 47A-352 |                    |     |     |
| 12pm - 1pm  |     |         |                    |     |     |

# Course Modules



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public holiday or semester break

Week 2

Weeks 3-8 (6 weeks)

Weeks 9-13 (4 weeks)

### Python basics

- Basic data structures
- Key functions and libraries

### Traffic assignment

- Graph theory, shortest path
- User Equilibrium (UE)
- System Optimisation (SO)
- Solution algorithms

### Discrete choice modelling

- Route choice/mode choice
- Applications (travel behaviour analysis, demand forecasting)

# Textbooks

## ► Traffic Assignment

- [BLU] Boyles, S. D., N. E. Lownes, and A. Unnikrishnan. (2025) *Transportation Network Analysis*, Volume I, Version 1.0.
  - The full text available for free at: <https://sboyles.github.io/blubook.html>.
- [Sheffi] Sheffi, Y. (1985) *Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods*
  - The full text available for free at: <https://sheffi.mit.edu/book/urban-transportation-networks>

## ► Discrete Choice Modelling

- [Train] Kenneth E. Train (2009) *Discrete Choice Methods with Simulation*, Second Edition
  - The full text available for free at: <https://eml.berkeley.edu/books/choice2.html>

# Assessment

## ► Summary

| Assessment Items                         | Weight      |
|------------------------------------------|-------------|
| In-class Quizzes                         | 20%         |
| Assignment 1 (Traffic Assignment)        | 15%         |
| Assignment 2 (Discrete Choice Modelling) | 15%         |
| Final Exam                               | 50%         |
|                                          | <b>100%</b> |

# Final Exam (50%)

- ▶ Reading: **10 minutes**, Exam duration: **120 minutes**
- ▶ **Closed book examination** – no written materials permitted
  - A formula sheet will be provided
- ▶ Short answer, short essay, and problem-solving questions
  - Showing problem solving steps and detailed calculations
  - Explaining concepts and definitions
- ▶ Past exam papers are available via the UQ Library website.
- ▶ **Hurdle:** A minimum of 40% (40 out of 100 marks) in the final exam is required to pass the course.

# Two Assignments (15% \* 2 = 30%)

|                                                 | Week 1<br>23-Feb-26 | Week 2<br>02-Mar-26 | Week 3<br>09-Mar-26 | Week 4<br>16-Mar-26 | Week 5<br>23-Mar-26 | Week 6<br>30-Mar-26 | Sem.Break<br>06-Apr-26 | Week 7<br>13-Apr-26 | Week 8<br>20-Apr-26 | Week 9<br>27-Apr-26 | Week 10<br>04-May-26 | Week 11<br>11-May-26 | Week 12<br>18-May-26 | Week 13<br>25-May-26 |
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| Assignments                                     |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |
| Two assignments,<br>each with a 2-week deadline |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |

| Assignment                               | Start date                    | Due date                      |
|------------------------------------------|-------------------------------|-------------------------------|
| Assignment 1 (Traffic Assignment)        | 5 PM, Tuesday, April 14, 2026 | 5 PM, Tuesday, April 28, 2026 |
| Assignment 2 (Discrete Choice Modelling) | 5 PM, Tuesday, May 12, 2026   | 5 PM, Wednesday, May 27, 2026 |

- ▶ **Two assignments** will be released in **Week 7** and **Week 11**, respectively.
  - Problem-solving questions, textbook problems, data analysis, Python programming
  - Assignments must be solved and submitted **individually**.
  - The standard **late submission penalty**: 10% penalty deducted per 24 hours for up to 7 days; 0 mark after 7 days
  - You may be able to [apply for an extension](#), which should be submitted **before the due date**.



# In-class Quizzes (20%)

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| In-class Quizzes |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |
| During Tutorials |                     |                     |                     |                     |                     |                     |                        |                     |                     |                     |                      |                      |                      |                      |

- ▶ **12 in-class quizzes** during tutorial sessions from **Week 2** to **Week 13**
  - To assess the understanding of the material covered in lectures and tutorials so far
  - To reinforce key concepts and analytical skills
- ▶ A quick quiz with **about 5–8 questions**, taking **10-15 minutes**
- ▶ **Multiple Choice Questions, Multiple Select Questions**
- ▶ No deferral or extension; 0 mark for the missed quizzes
- ▶ The **best 10 of the 12 quizzes** will count toward the final grade.
- ▶ **Each of these quizzes is worth 2%**, making up a total quiz weight of **20%**.

# In-Class Quiz Participation

► We will use **EchoPoll**, a live polling tool available at UQ.

**Submit answer via mobile phone**

**A question slide displayed in class**

**What is the capital of Australia?**

- A. Sydney
- B. Melbourne
- C. Canberra
- D. Brisbane

60

Multiple Choice 0 POINTS

What is the capital of Australia?

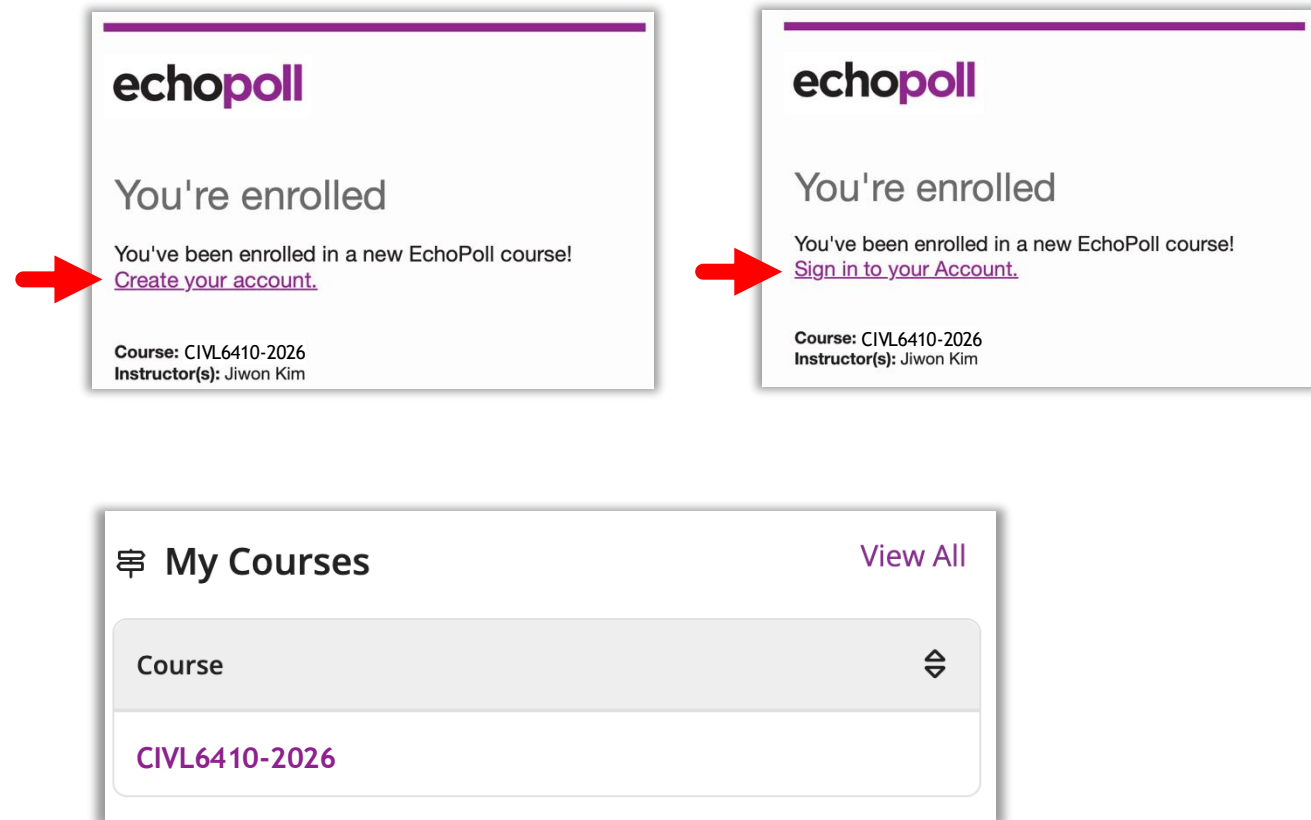
- A
- B
- C
- D

Select a response

CHAT LEADERBOARD HIDE MEDIA EXIT

# In-Class Quiz Participation

- ▶ To join sessions, you need to be enrolled in the **EchoPoll course** for CIVL6410.
- ▶ An invitation email to join the EchoPoll Course ('**CIVL6410-2026**') has been sent to **your UQ student email** (sXXXXXXX@student.uq.edu.au).
- ▶ Click the link in the email to either create a new EchoPoll account or sign in to your existing account.
- ▶ Once logged in to your EchoPoll account, you will see **CIVL6410-2026** in your course list.



# In-Class Quiz Participation

## Participation Rules

### ► Mobile Phones Only

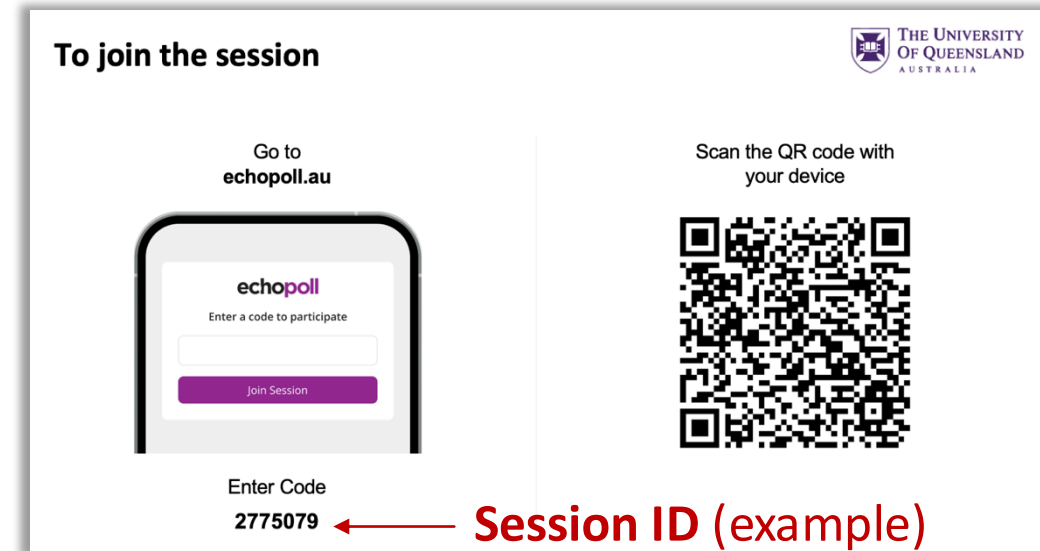
- Laptop or tablet screens are large and highly visible, which can unintentionally expose answers to nearby students.

### ► Keep Your Answers Private

- Hold your phone in a way that prevents others from viewing your screen.
- Do not show, share, or discuss answers during the quiz.

### ► No External Assistance

- No internet searching, AI tools, note viewing, messaging, or photo taking during the quiz.

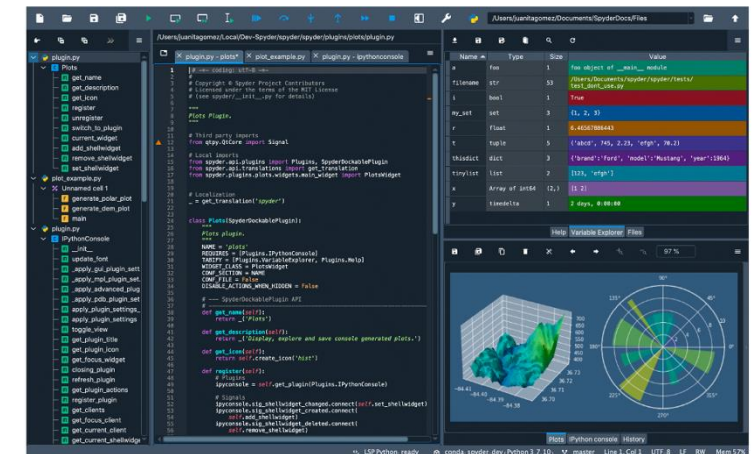


# Python Programming

- ▶ **Python Programming** is used extensively throughout the semester for **tutorials** and **assignments**.
- ▶ **Bring your own device (BYOD)**
  - <https://my.uq.edu.au/information-and-services/information-technology/bring-your-own-device-byod>
  - [Laptop loans](#) are also available from the library.

# Python Programming

- ▶ We will use '**Anaconda**' to install packages and tools for Python programming.
  - 'Anaconda' is an open-source distribution of Python programming languages. It is designed to simplify the process of managing and deploying packages and libraries commonly used in scientific computing, data analysis, and visualisation.
- ▶ We will use the '**Spyder**' IDE for writing, running, and debugging Python codes.
  - 'Spyder' is included in the **Anaconda** distribution
- ▶ **Complete 'Setting Up Python Development Environment' prior to Week 2 lecture.**
  - Blackboard > Content > Python Programming > CIVL6410\_python\_setup.pdf



# Python Programming

► A recommended resource for Python basics:

➤ <http://avinashu.com/tutorial/indexPythonlong.html>

➤ Go through the **Python Tutorials** provided there:

## Python Tutorial

- Basics of Python: [HTML Notebook](#)
- Built-in Data Structures: [HTML Notebook](#)
- Functions and Classes: [HTML Notebook](#)
- Input and Output: [HTML Notebook](#)
- Numpy: [HTML Notebook](#)
- Plots: [HTML Notebook](#)
- Scipy: [HTML Notebook](#)
- NetworkX: [HTML Notebook](#)
- Gurobi: [HTML](#)

## Introduction to Python

*Avinash Unnikrishnan (Portland State University, <http://avinashu.com/>)*

*Stephen Boyles (The University of Texas Austin, <https://sboyles.github.io/>)*

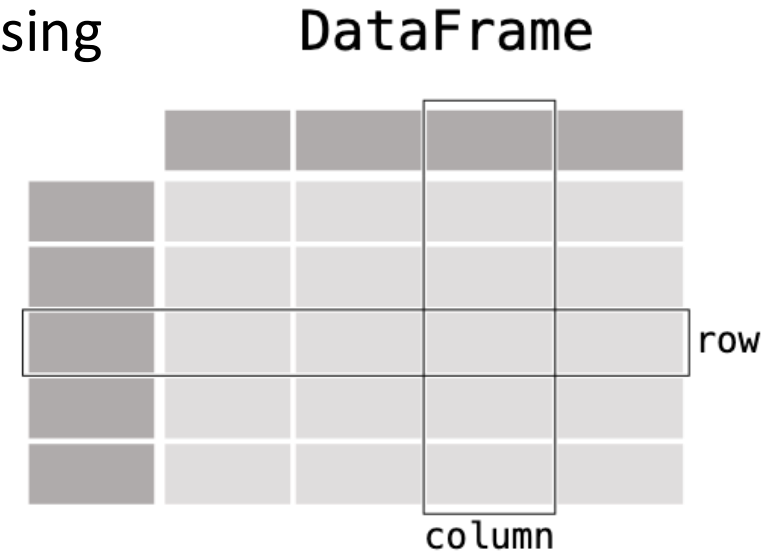
## Overview

The goal of this tutorial is to give a concise introduction to Python and is aimed at Civil and Environmental Engineering undergraduate and graduate students. This tutorial is by no means comprehensive. There are several very good detailed sources for Python. Some of the popular Python resources are listed below:

Created by the authors of the BLU book

# Python Programming - pandas

- ▶ **'Pandas'** is a widely used Python library for storing and analysing tabular data using a data structure called a **'DataFrame'**.
- ▶ Tutorials
  - Pandas official **'Getting Started'** tutorials:  
[https://pandas.pydata.org/docs/getting\\_started/index.html](https://pandas.pydata.org/docs/getting_started/index.html)
  - Additional resource: UIUC's **'Guides Using pandas DataFrames'**:  
<https://discovery.cs.illinois.edu/guides/>
- ▶ API (Application Programming Interface) Reference
  - Functions, methods, and attributes associated 'pandas.DataFrame' class:  
<https://pandas.pydata.org/docs/reference/frame.html>





# Appropriate Use of Artificial Intelligence (AI) Tools

Course Profile Statement:

***“Artificial Intelligence (AI) and Machine Translation (MT) are emerging tools that may support students in completing this assessment task. Students may **appropriately use** AI and/or MT in completing this assessment task. Students must clearly reference any use of AI or MT in each instance.”***

## ► In-Class Quizzes (Strictly Closed Resources)

- Quizzes are designed to assess your **individual understanding and independent problem-solving ability**. Any use of AI or external assistance during quizzes constitutes academic misconduct.

## ► During in-class quizzes:

- **No internet search** is permitted.
- **No digital or physical notes** are permitted.
- **No AI tools of any kind** are permitted.
- **No machine translation tools** are permitted.
- **No photo taking** is permitted.

# Appropriate Use of Artificial Intelligence (AI) Tools

Course Profile Statement:

***“Artificial Intelligence (AI) and Machine Translation (MT) are emerging tools that may support students in completing this assessment task. Students may **appropriately use** AI and/or MT in completing this assessment task. Students must clearly reference any use of AI or MT in each instance.”***

## ► Take-Home Assignments (Limited and Transparent Use Allowed)

- For take-home assignments, AI may be used **only in supportive ways**, not as a substitute for your own thinking.
- In completing Assignments, students must **NOT** use AI to:
  - Produce content that replaces original thought or reasoning
  - Interpret data, findings, or outputs
  - Produce final analysis results, discussion sections, conclusions, or insights
  - Generate complete solutions for submission
  - Generate code that is copied and pasted without understanding

# Appropriate Use of Artificial Intelligence (AI) Tools

- ▶ If you use AI to do your **work for you**, you'll be **replaced by it**; if you use AI to master the knowledge and skills needed to **do the work yourself**, you'll be the one **leading it**.
- ▶ **Building knowledge is like building muscle** – you can't ask a coach to lift the weights for you. **Only by doing the hard training yourself**, you do build real strength.
- ▶ **When you always rely on AI at your fingertips**, you risk becoming helpless without it — in the real world, the most important decisions and influence happen **face-to-face**, in moments where you cannot pause to consult a tool.

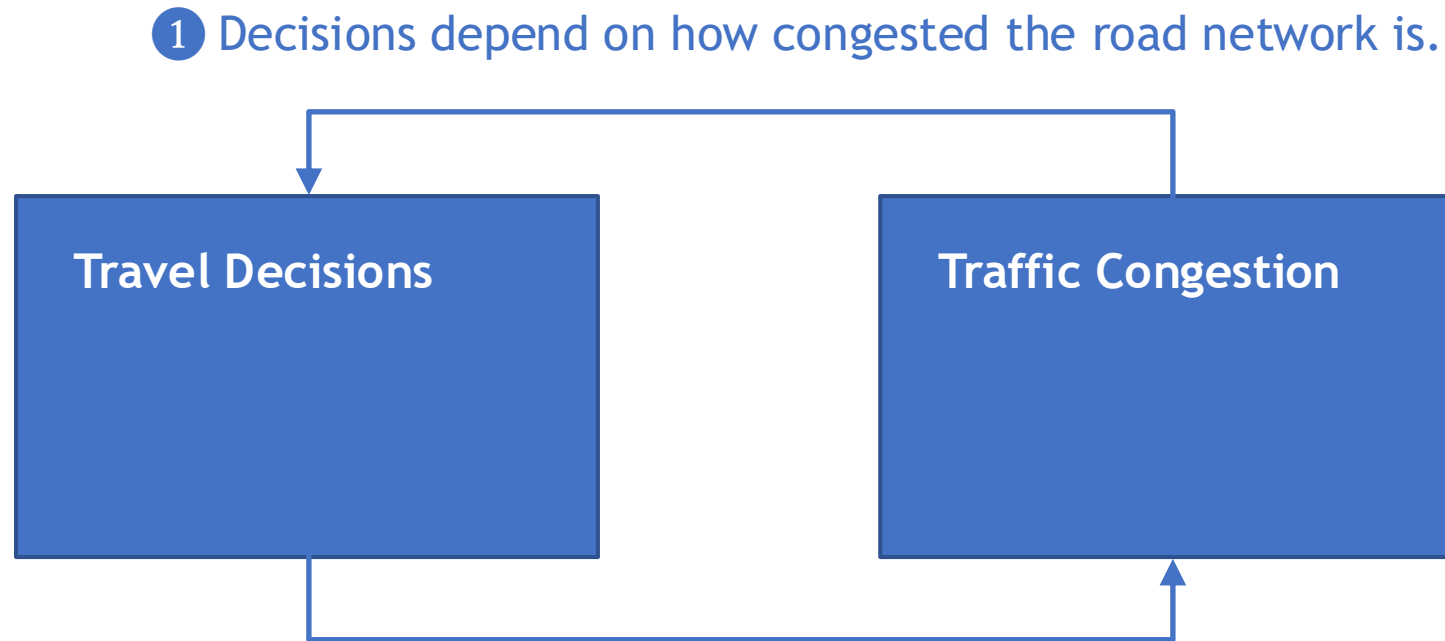
Think of **knowledge and thinking skills** as **freedom**: when you've built them yourself, you can speak, respond, and influence others confidently **with your own mind**.

If you rely entirely on AI, you become like **a fish in water** — functioning only when connected to AI, but **never truly free on your own**.

# Course Overview

# Transportation Network Modelling

- ▶ Aims to model interactions between **travel decisions** and **traffic congestion**.

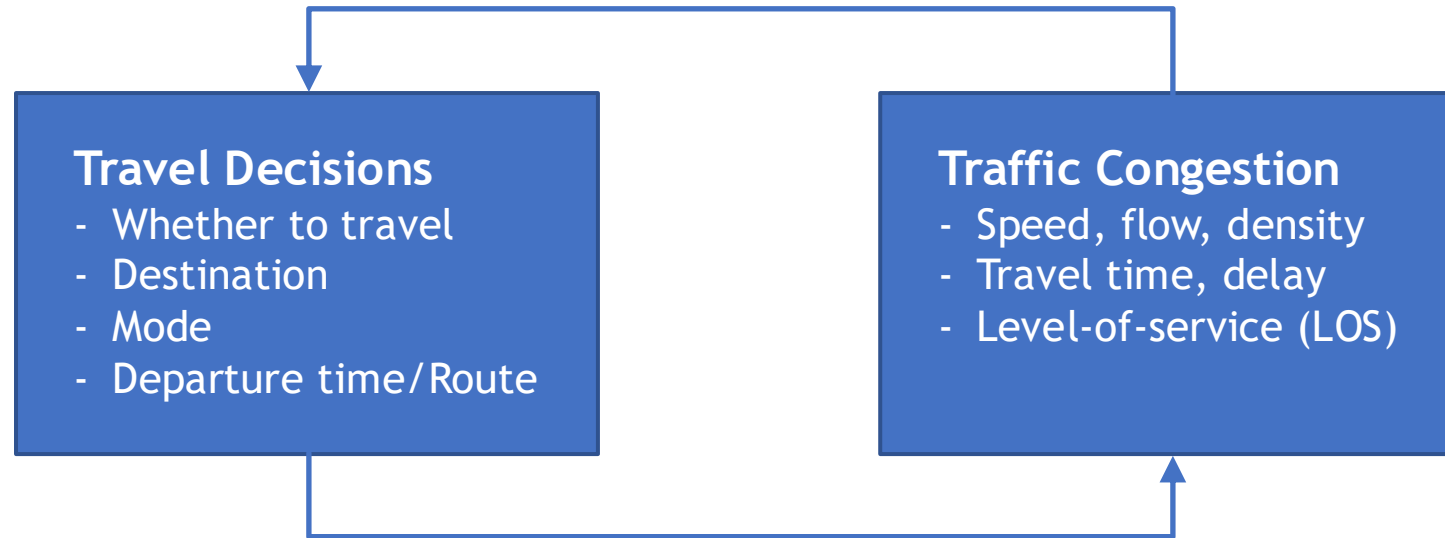


- ② Traffic congestion at any location in the network depends on how many people choose to travel through that location.

# Transportation Network Modelling

- Aims to model interactions between **travel decisions** and **traffic congestion**.

① Decisions depend on how congested the road network is.

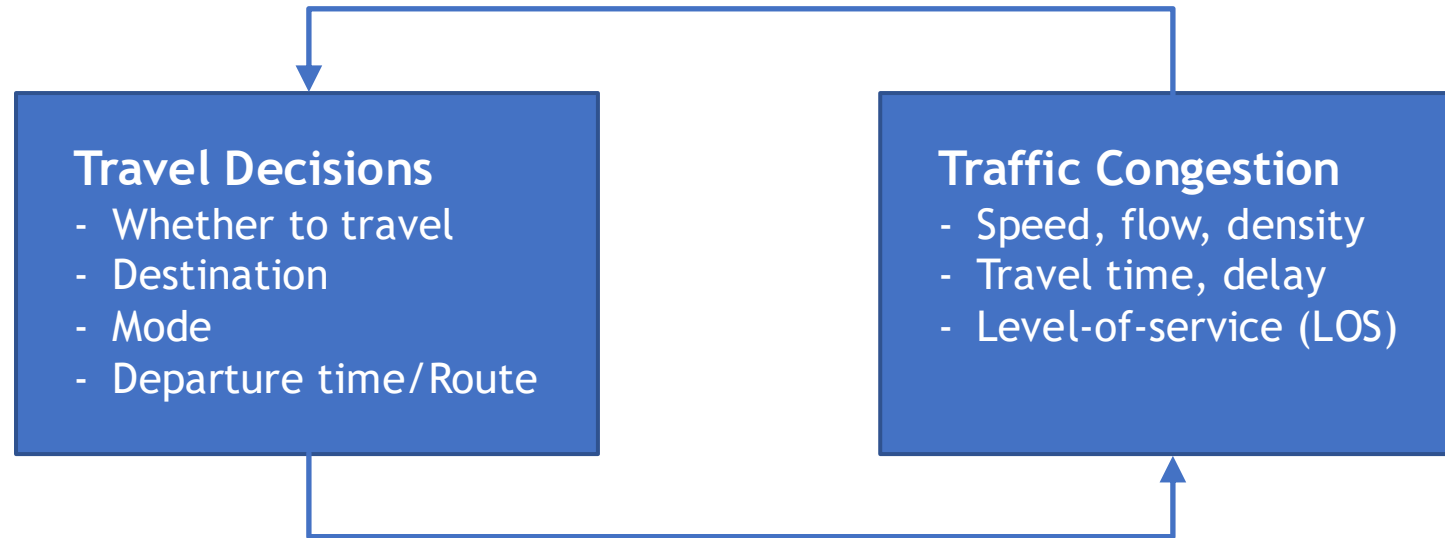


② Traffic congestion at any location in the network depends on how many people choose to travel through that location.

# Transportation Network Modelling

- Solves ❶ and ❷ simultaneously to obtain the **flow pattern** throughout the network.

❶ Decisions depend on how congested the road network is.



❷ Traffic congestion at any location in the network depends on how many people choose to travel through that location.

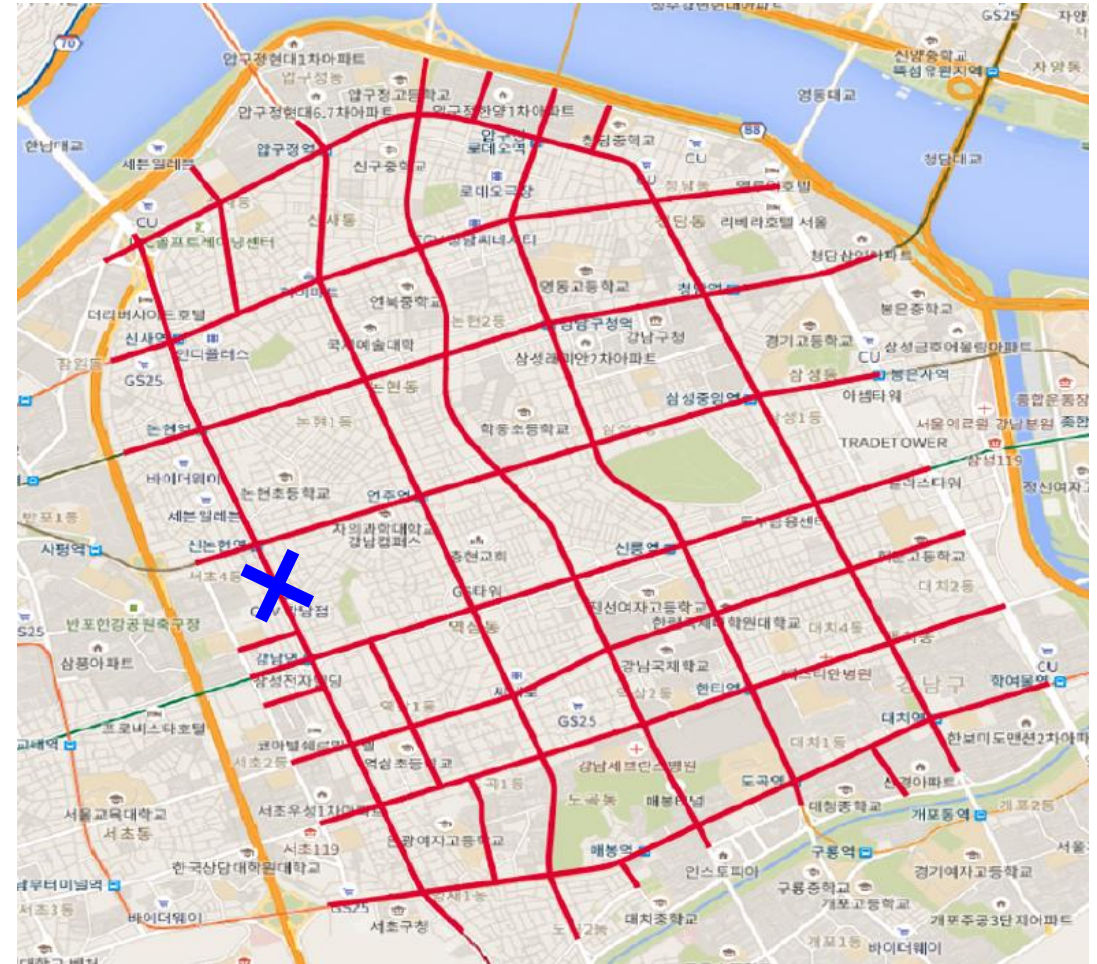
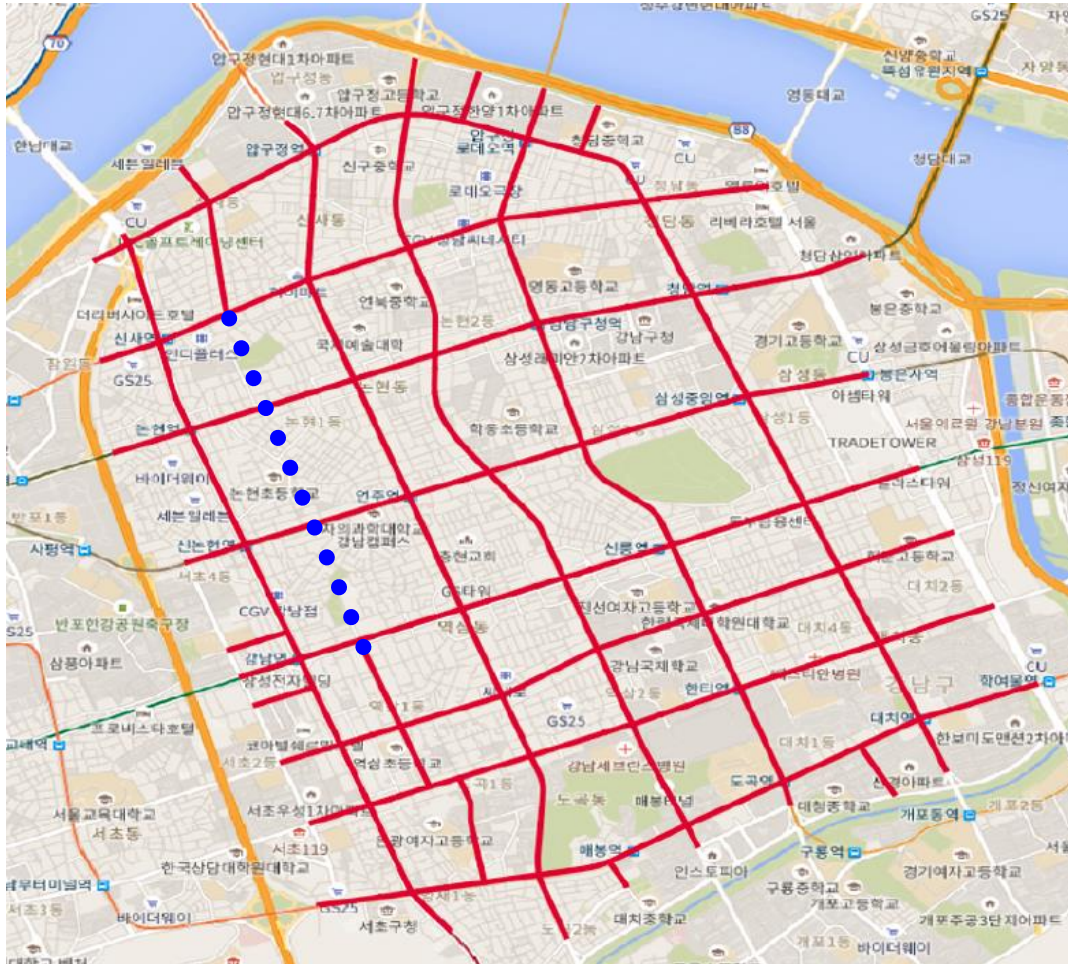
# Purpose of Transportation Network Modelling

- ▶ To predict the impact of given transportation scenarios on an urban transport network
  - How would people react to specific changes in infrastructure, services, and traffic control policies and what would be the resulting congestion levels across the network?
    - Build a new freeway, a new transit line, etc.
    - Deploy new traffic control and operation strategies (e.g., traffic signal, traveller information, etc.)
    - Deploy new demand management strategies (e.g., toll, incentives, etc.)
- ▶ To evaluate alternatives, projects, or policies
  - Support transportation infrastructure investment decisions



## Example

### ► Adding a new road or Closing a road section

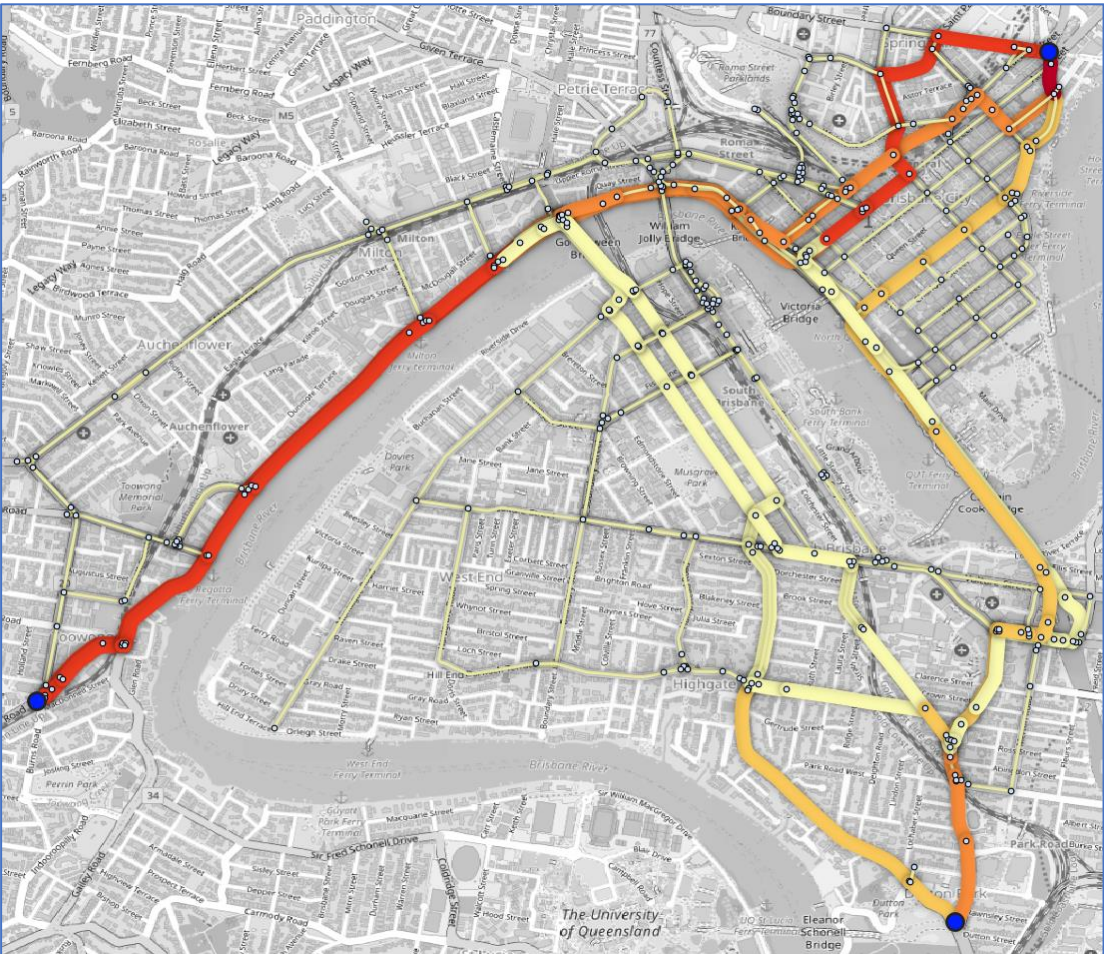


Gangnam, South Korea

# Example

## ► Constructing a new bridge

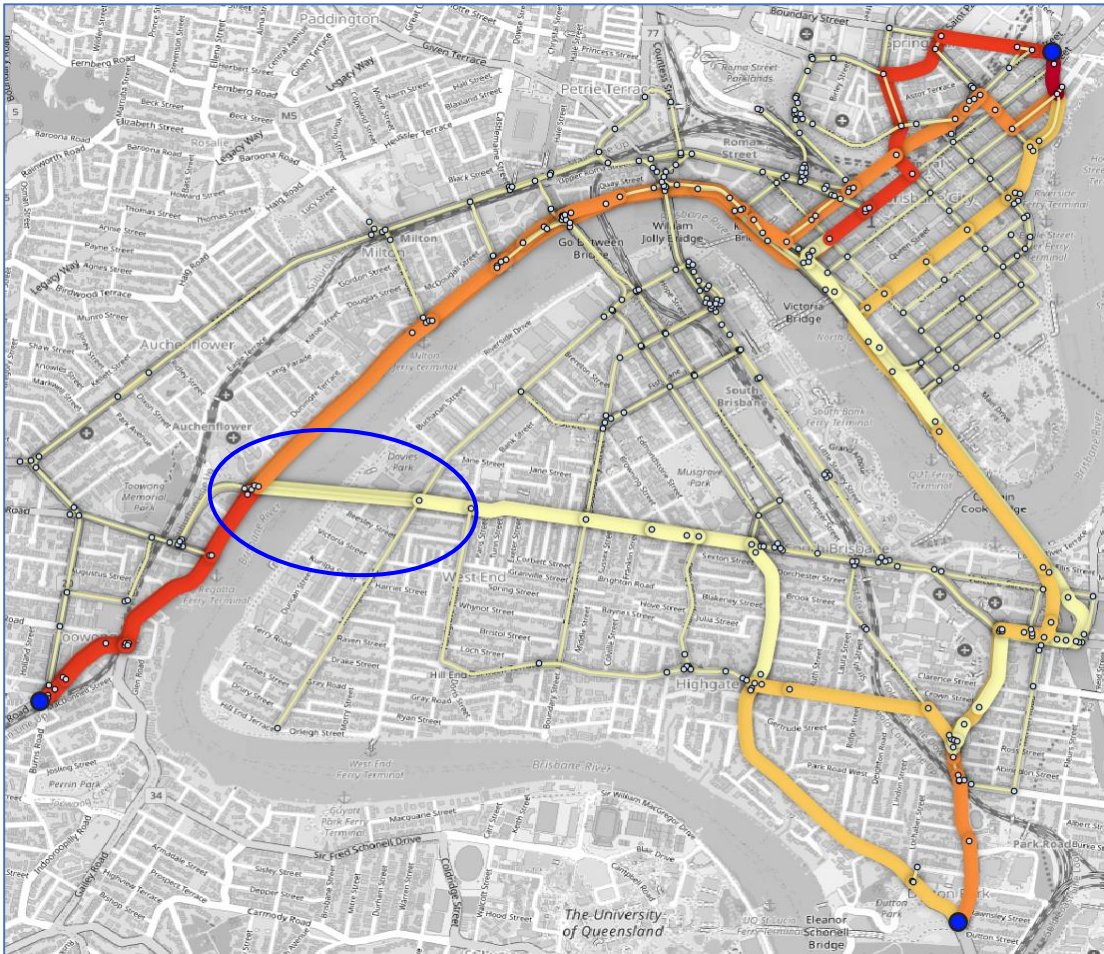
Original Network



Brisbane

Modified Network

with new links connecting *Coronation Drive* and *West End*





## Example

- Switching an intersection from a roundabout to a signalised intersection

November 2013



April 2016



Sir Fred Schonell Dr / Sir William Macgregor Dr / Campbell Rd / Mill Rd intersection at UQ

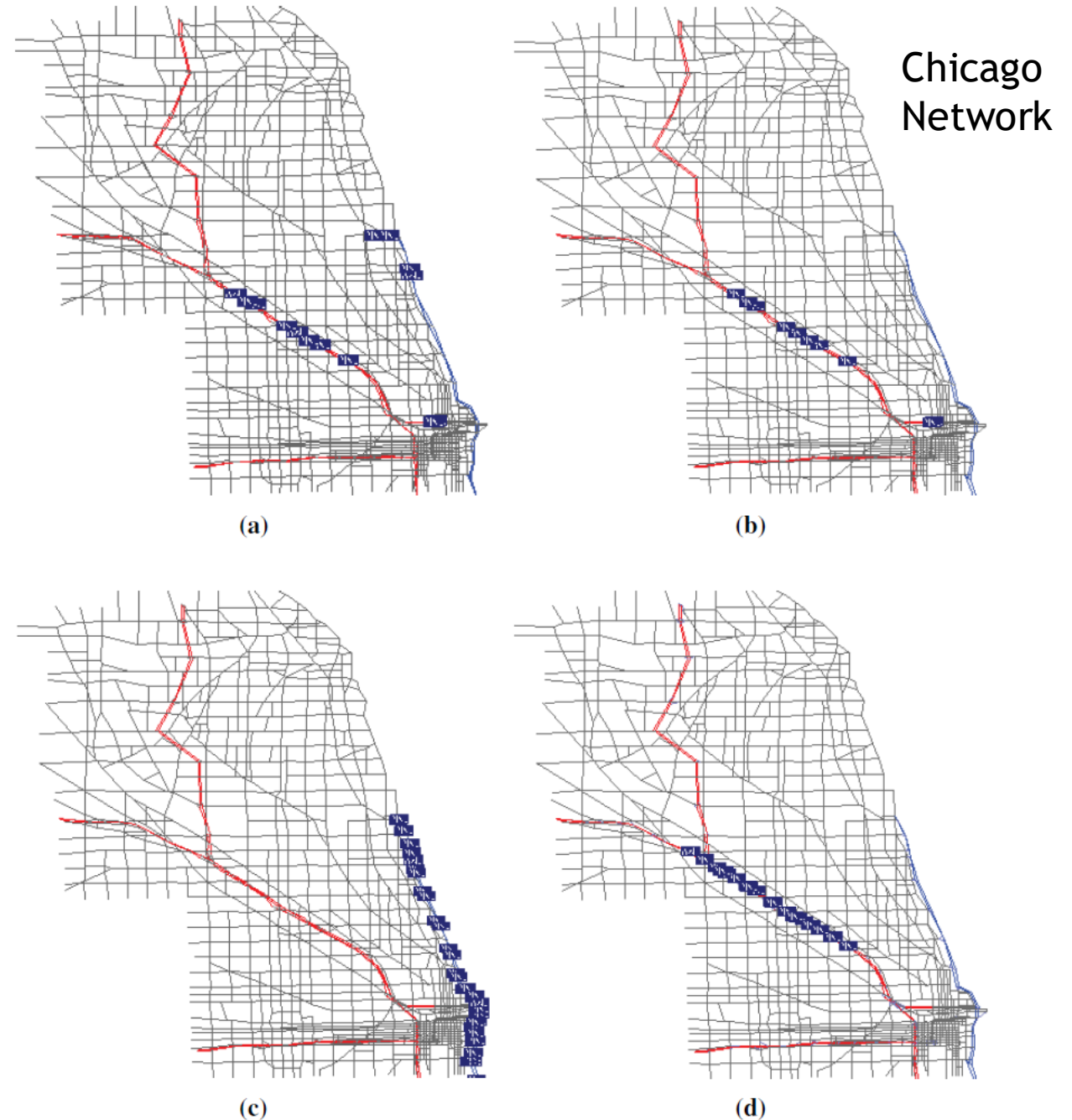
## Example

### ► Four scenarios for Variable Message Sign (VMS) strategies

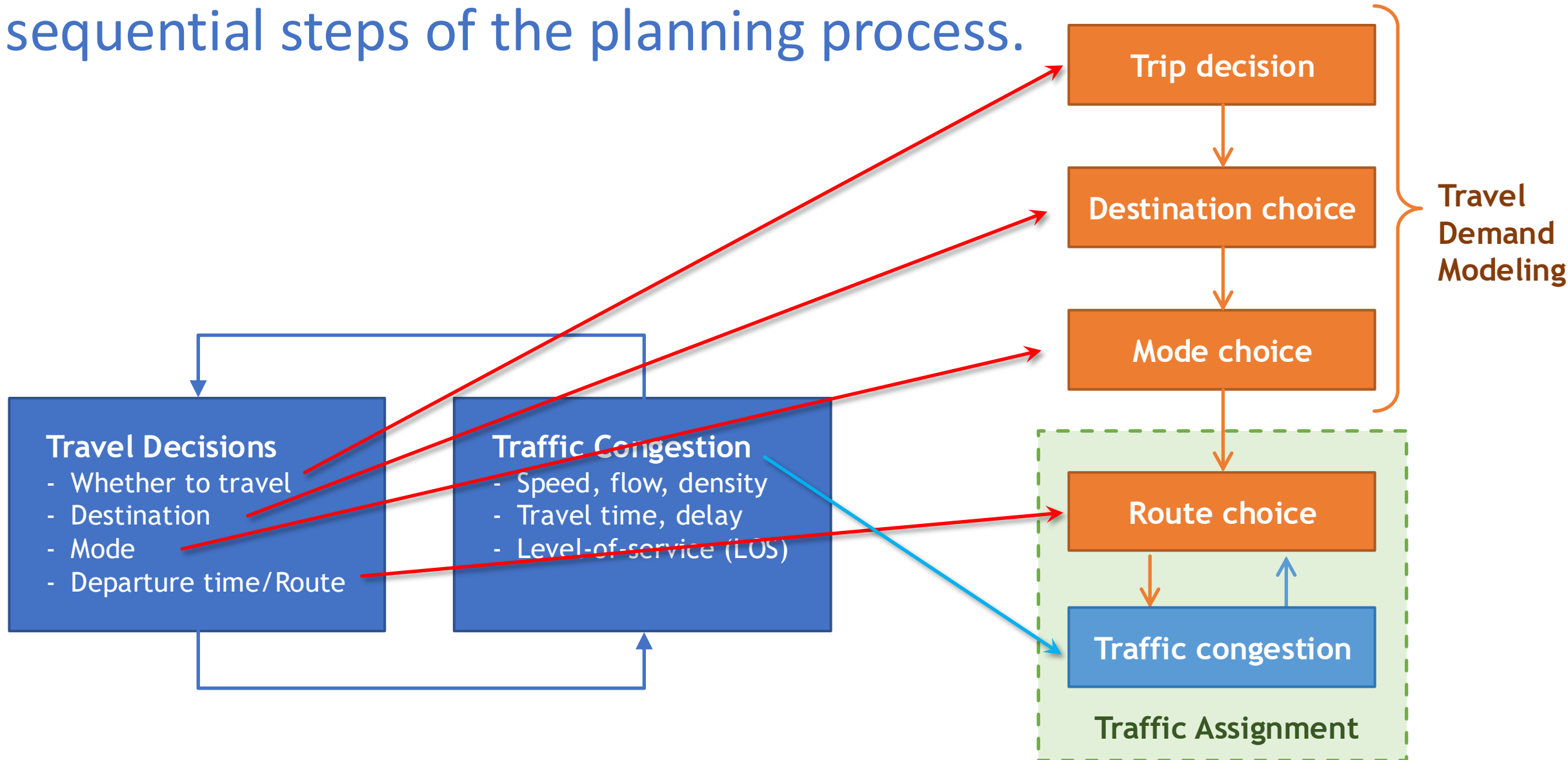
- (a) Advisory VMSs on Kennedy Expressway and Lake Shore Drive
- (b) Advisory VMSs on Kennedy Expressway only
- (c) Control VMSs on Lake Shore Drive
- (d) Control VMSs on Kennedy Expressway

**Advisory VMSs** display the traffic information (e.g., traffic congestion ahead) and suggested detour paths.

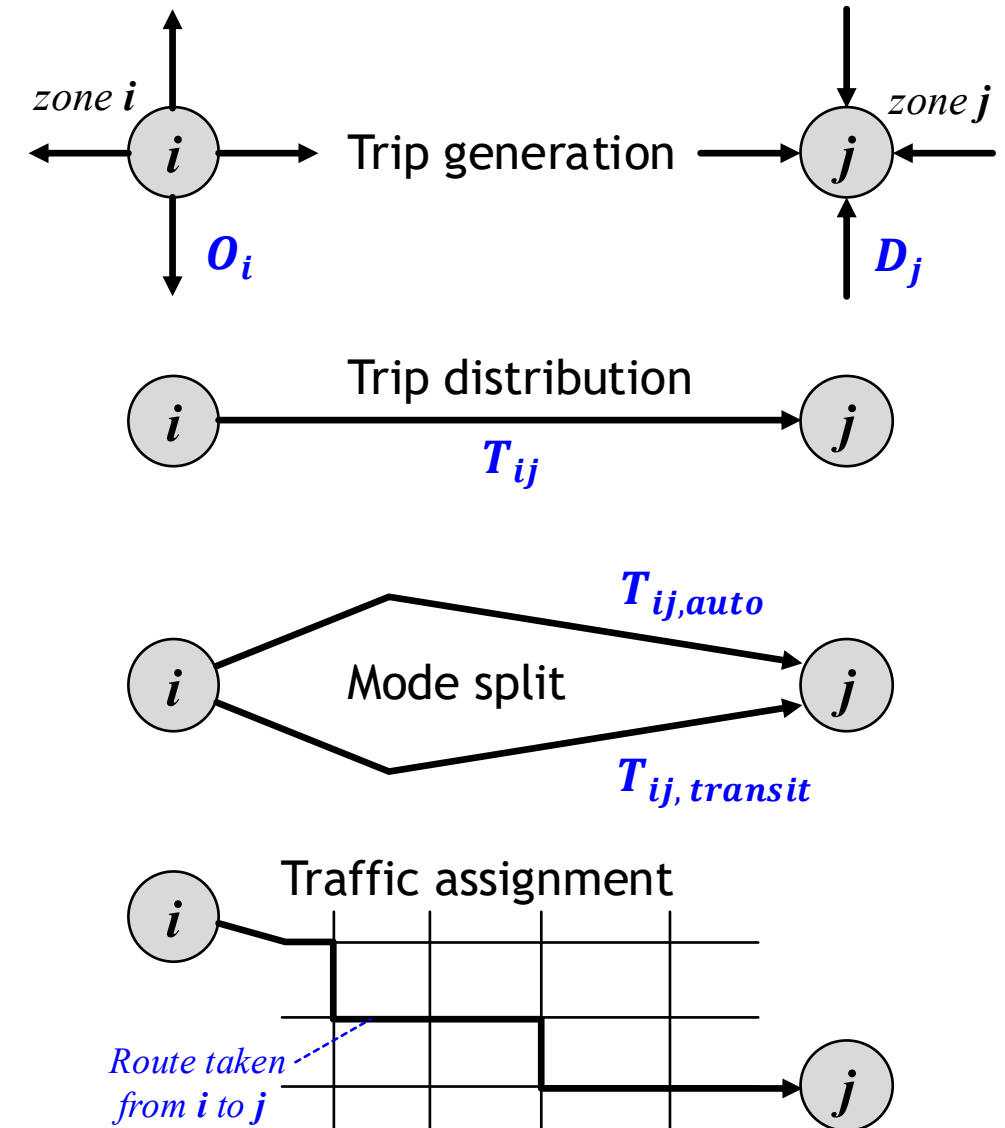
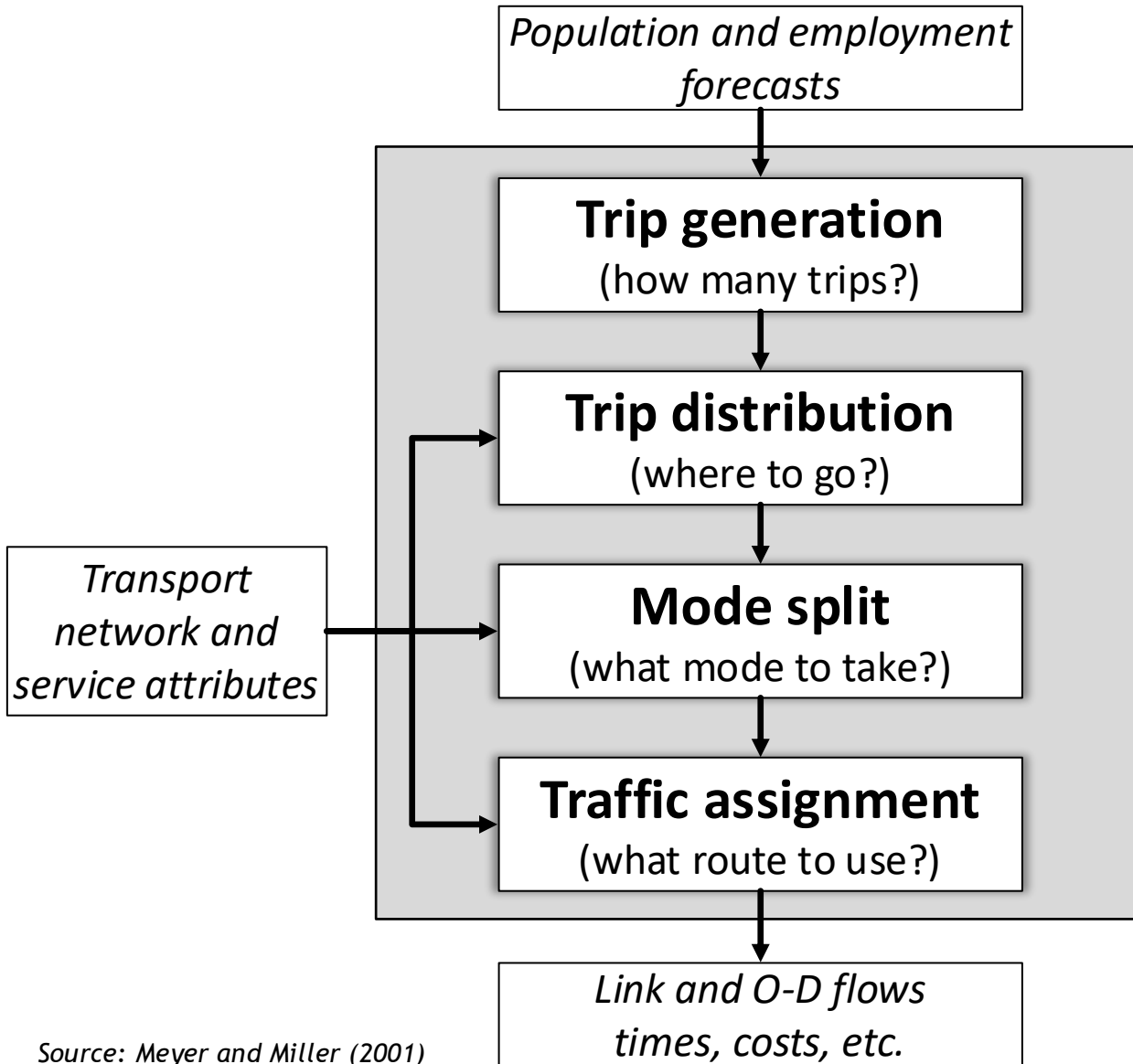
**Control VMSs** display variable speed limits (VSL).



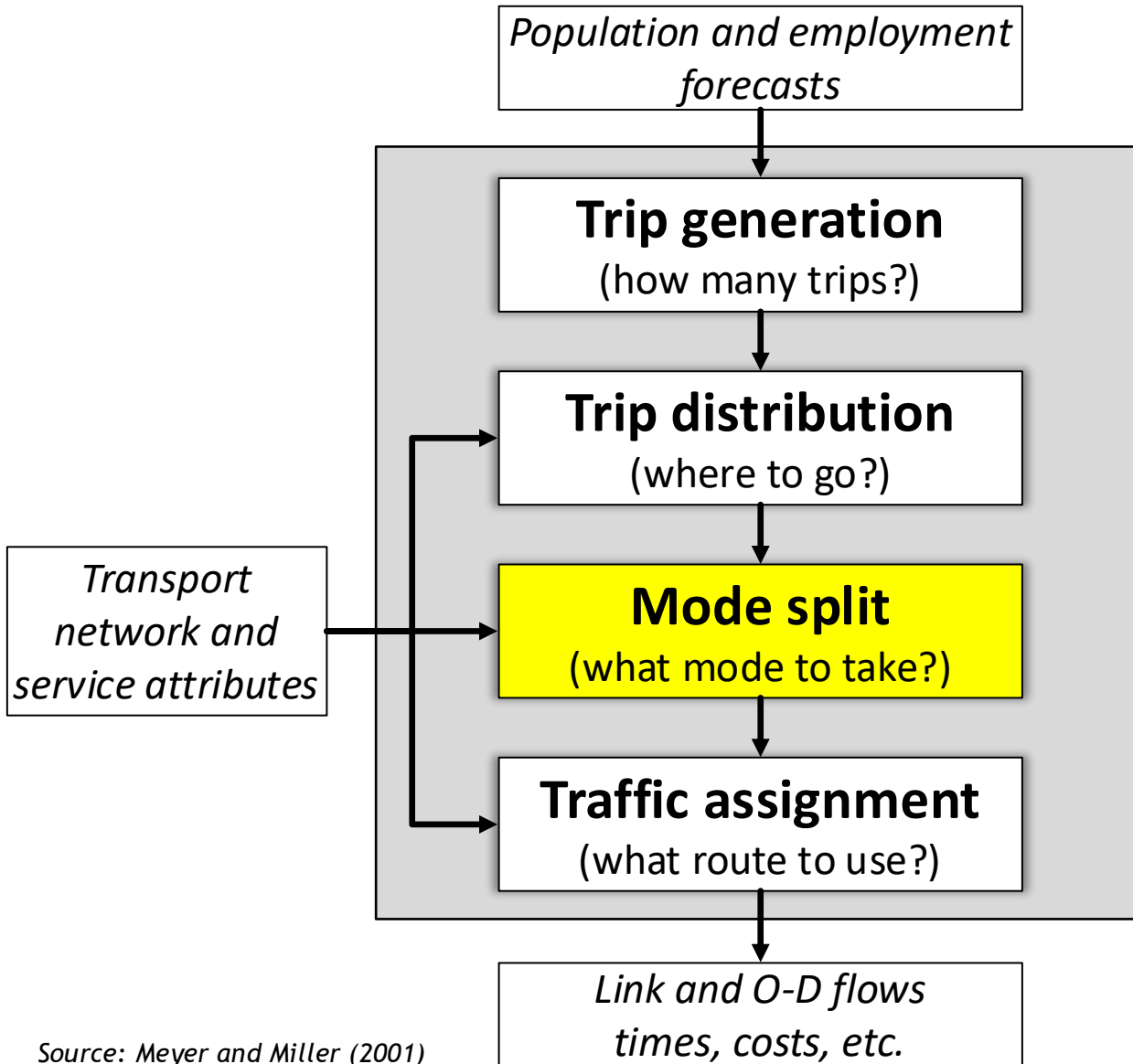
Transport network modelling traditionally forms the sequential steps of the planning process.



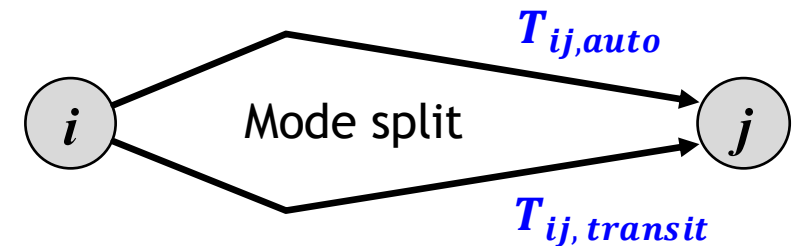
# Traditional 4-Step Model



# The Mode Choice problem

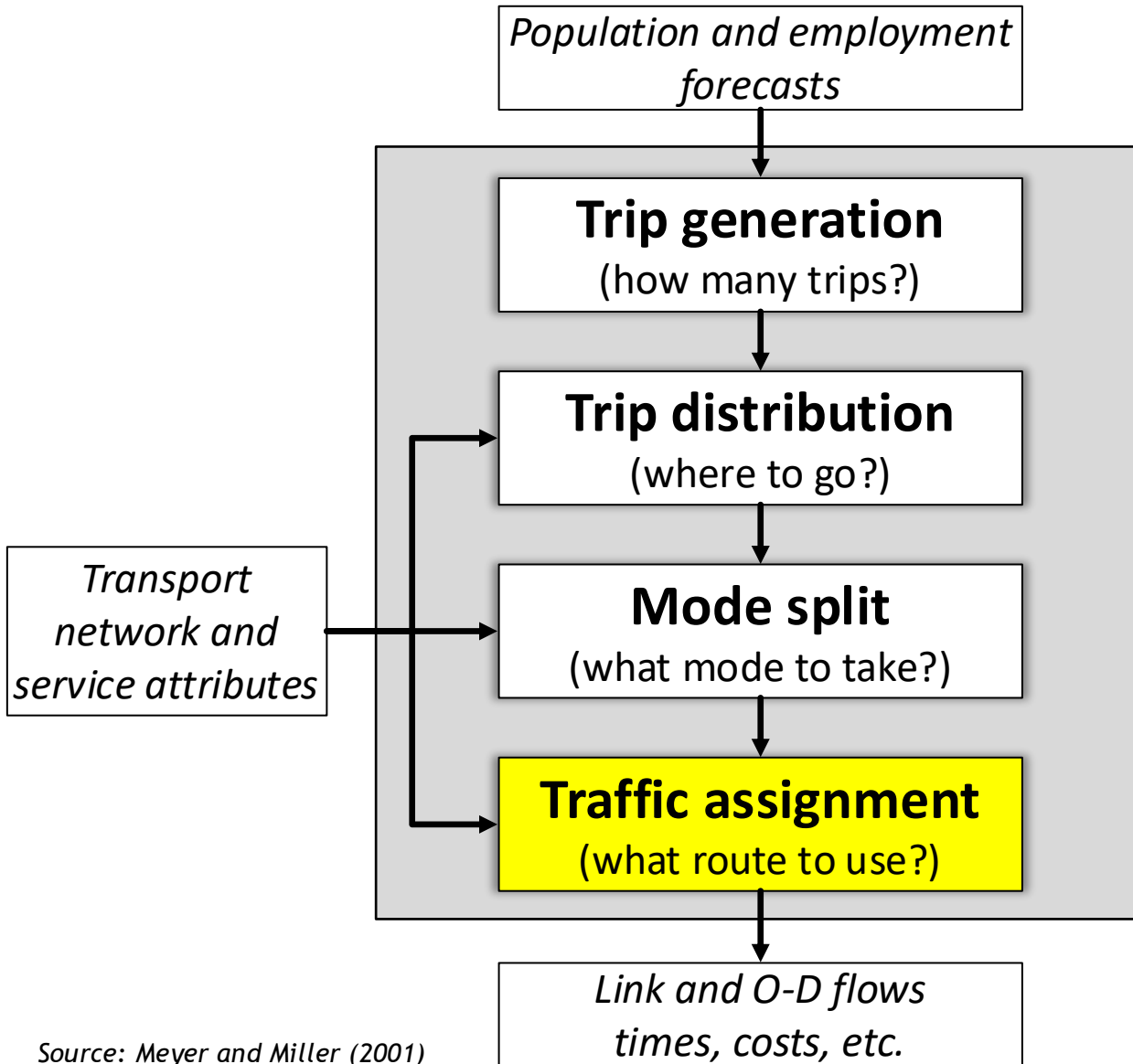


- ▶ How do travellers choose transport modes (car, bus, train, walk, bicycle, etc.)?
- ▶ What influences travellers' choice behaviours?

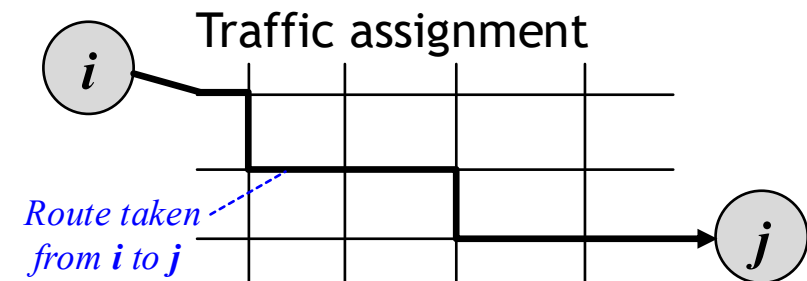




# The Traffic Assignment problem



- ▶ How are travelers assigned to various **routes** on the network, given the known **travel demand input**:
- Origin-Destination matrix or OD matrix, containing the aggregated flows between each origin-destination pair





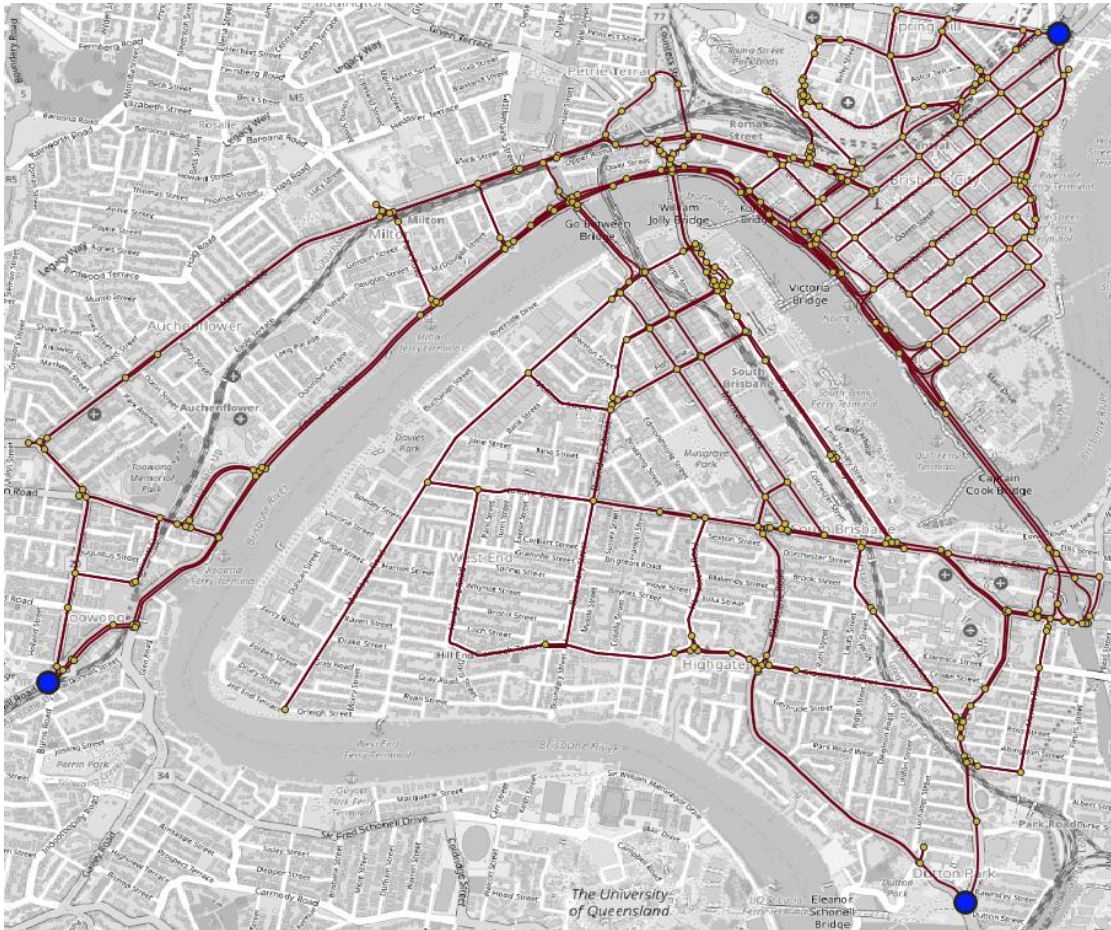
# Traffic Assignment

# Representation of the Transportation Network

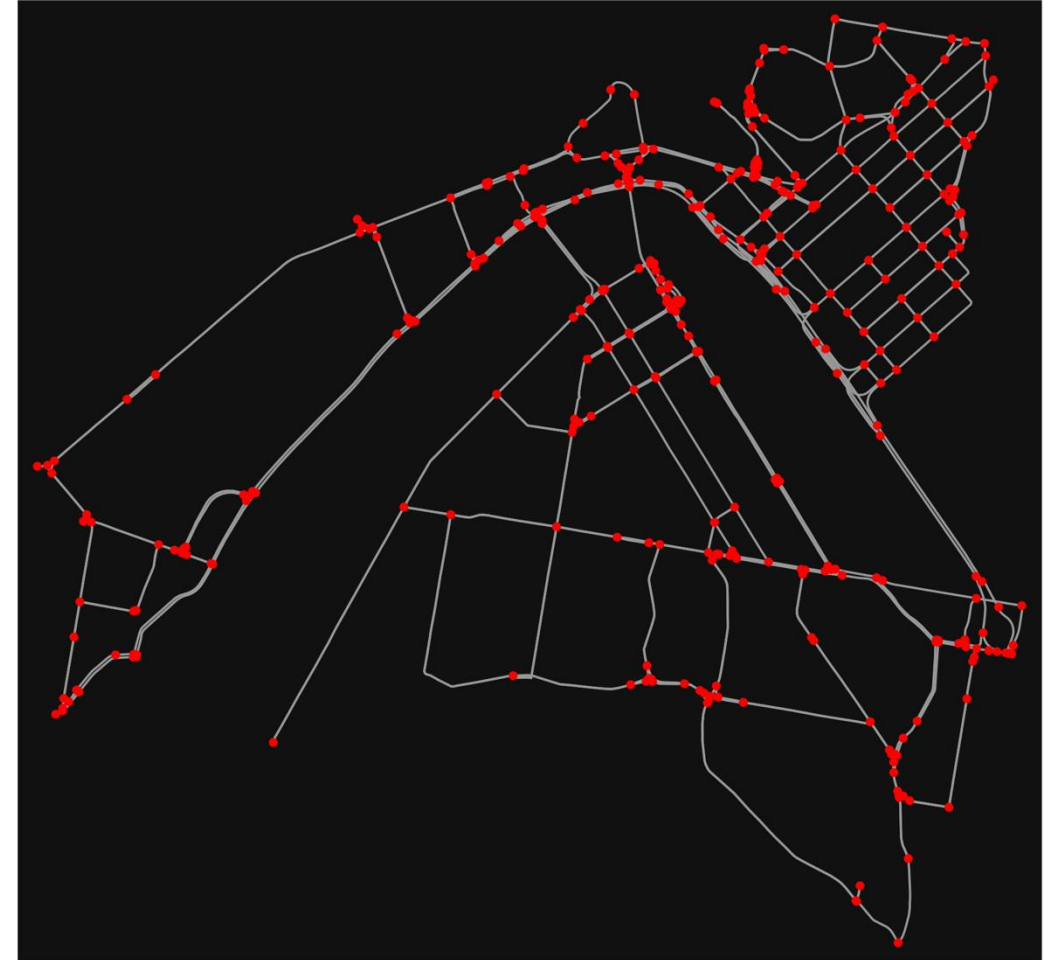
- ▶ For mathematical modelling purposes, a transportation system is represented as a **network**, consisting of **nodes** and **links**.
  - A **link** represents a means of travel from one point to another
    - e.g., A road segment between two intersections; a bus route between two stops, etc.
  - The **nodes** are the end points of the links

# Representation of the Transportation Network

Physical road network



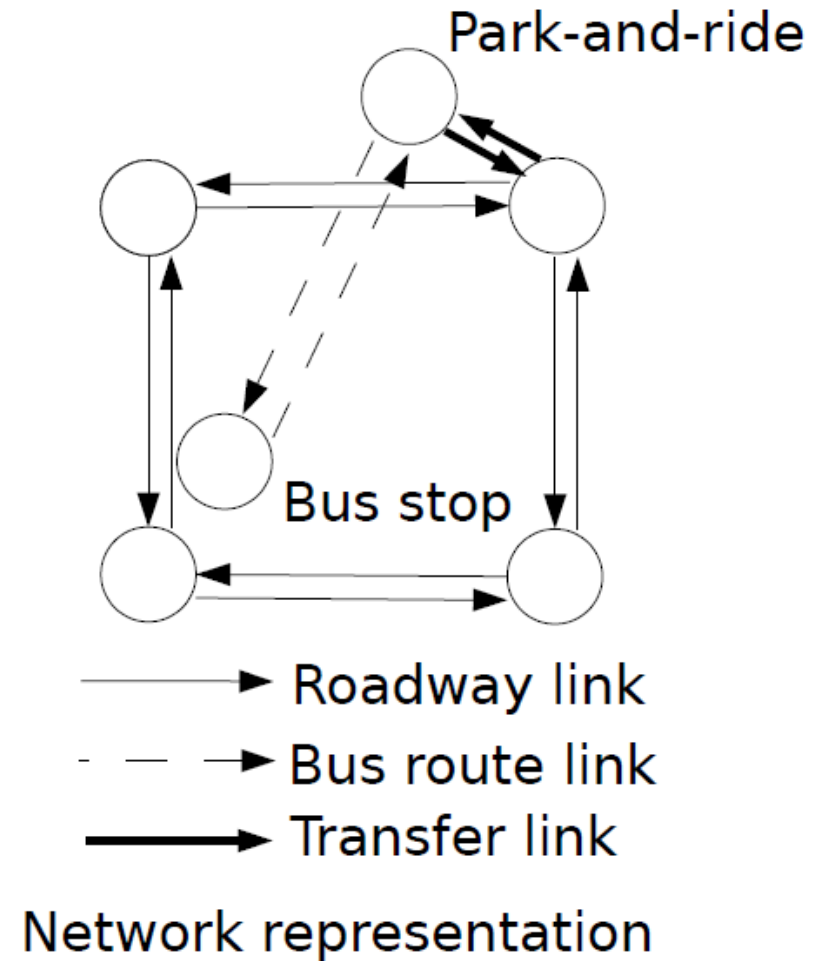
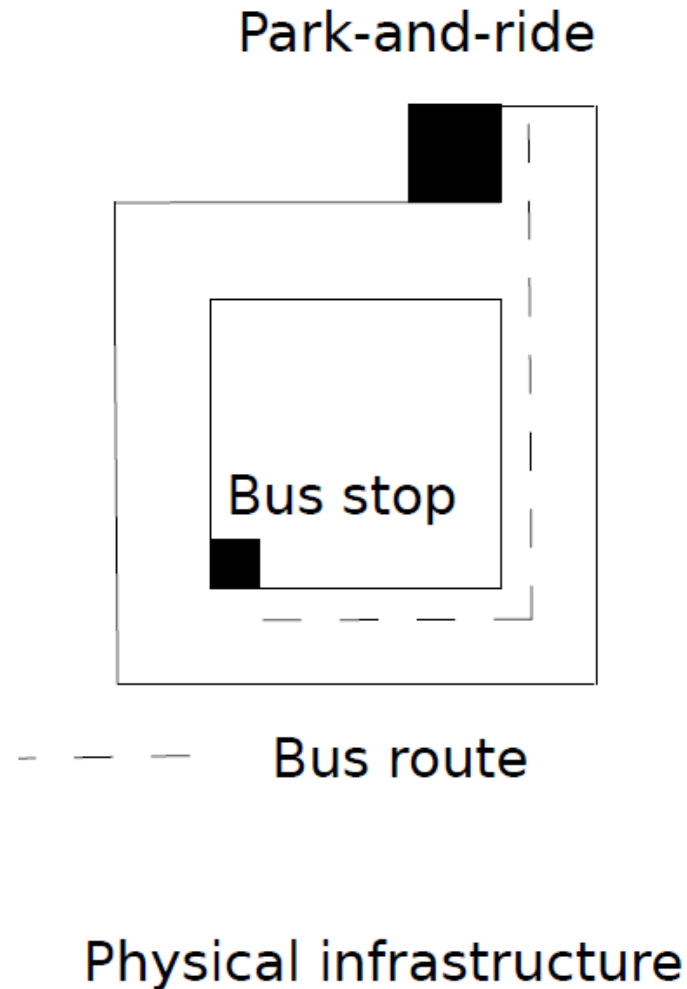
Network (Graph) Representation



Brisbane

# Representation of the Transportation Network

## ► Example



# Representation of the Transportation Network

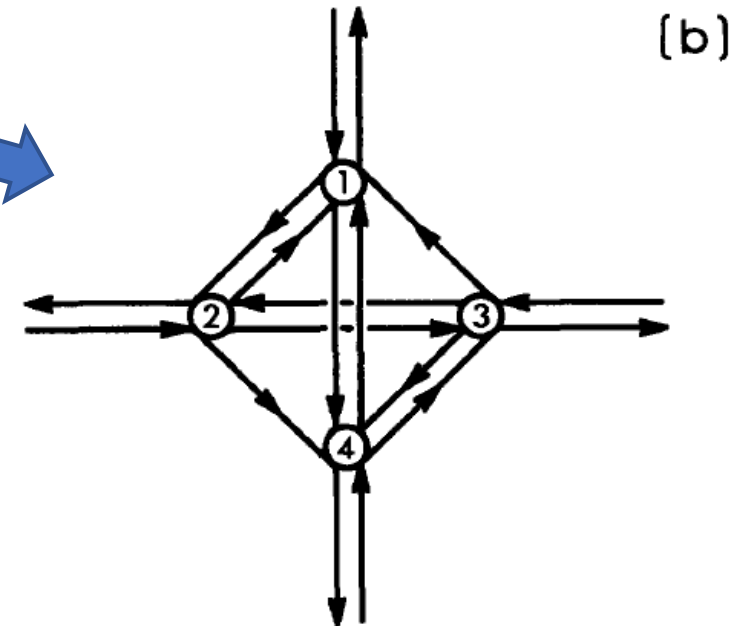
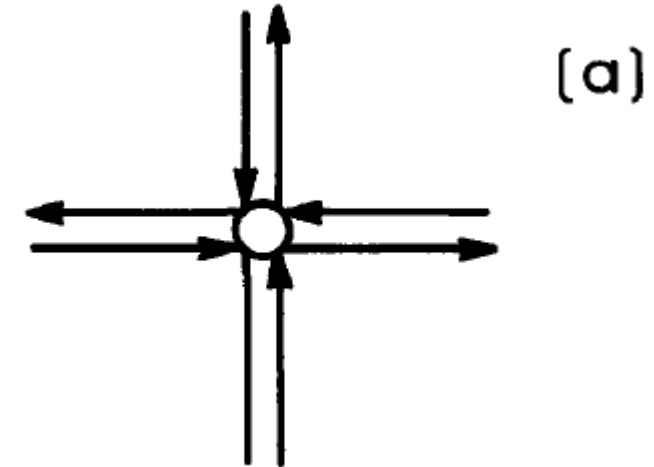
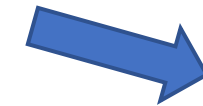
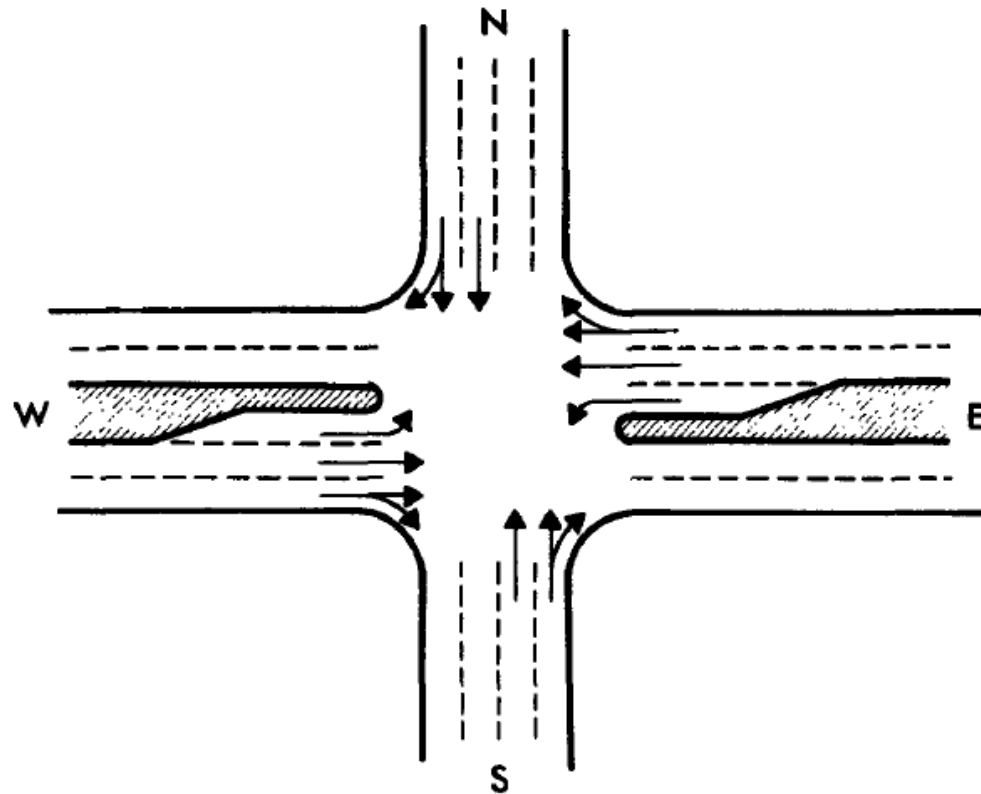
► Example

Table 1.1: Nodes and links in different kinds of transportation networks.

| Network type   | Nodes                            | Links             |
|----------------|----------------------------------|-------------------|
| Roadway        | Intersections                    | Street segments   |
| Public transit | Bus or train stops               | Route segments    |
| Freight        | Factories, warehouses, retailers | Shipping options  |
| Air            | Airports                         | Flights           |
| Maritime       | Ports                            | Shipping channels |

# Representation of the Transportation Network

- ▶ Two representations of the same intersection
  - The level of detail in a network varies with applications.



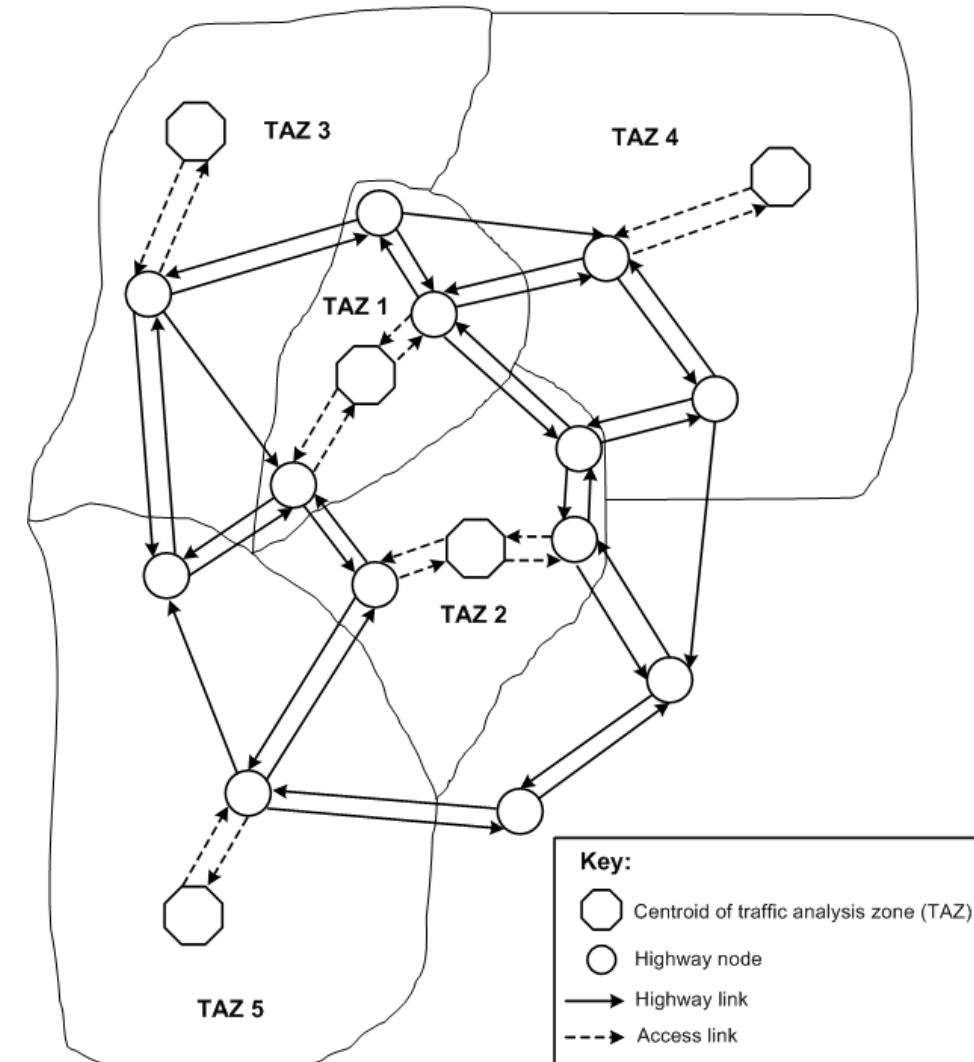
# Zones and Centroids

## ► Traffic Analysis Zone (TAZ)

- The process of transport network modelling starts with segmenting the geographic region being studied into *nearly homogenous areas* called **TAZs** based on similarities in land use, socioeconomic conditions, etc.

## ► Number of TAZs

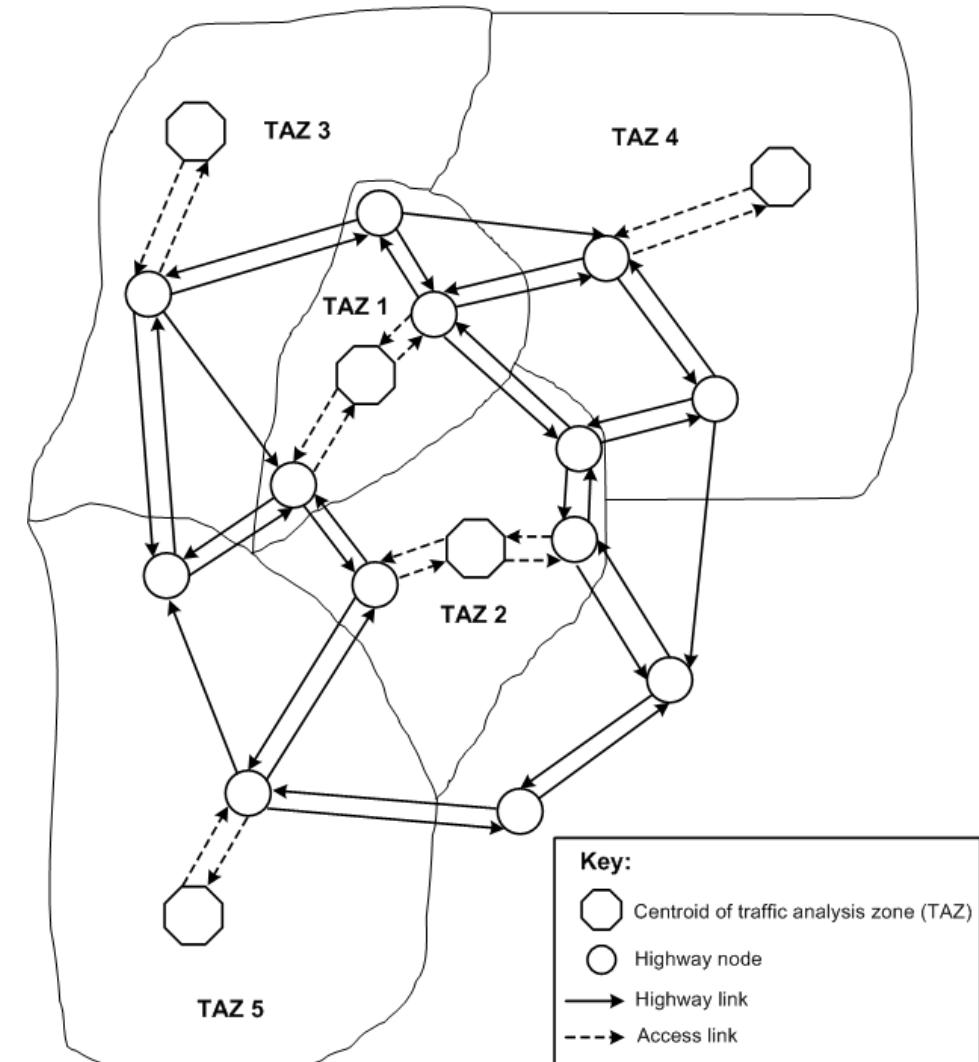
- Trade-off between **accuracy** (smaller TAZs provide more detailed forecasts) and **ease of implementation** (larger TAZs require fewer data and less computation)





# Zones and Centroids

- ▶ Centroid of TAZ
  - Each zone is represented by a node known as a *centroid*.
    - **Source** nodes = *origin* zones
    - **Sink** nodes = *destination* zones
- ▶ **Origin-Destination (OD) matrix** specifies the flow between every origin centroid and every destination centroid in the network.

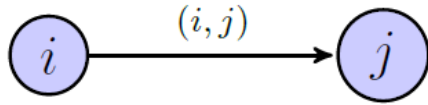




# Network and Link

- ▶ A **network** is denoted as  $G = (N, A)$ , where  $N$  is the set of nodes and  $A$  is the set of links.

- ▶ A **link** connecting node  $i$  and node  $j$  is denoted as  $(i, j)$ :



- ▶ Node  $i$  is called the '*tail*' or 'upstream node'
  - ▶ Node  $j$  is called the '*head*' or '*downstream node*'
- ▶ **Example:**
    - ▶  $N = \{1, 2, 3, 4\}$
    - ▶  $A = \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 4)\}$

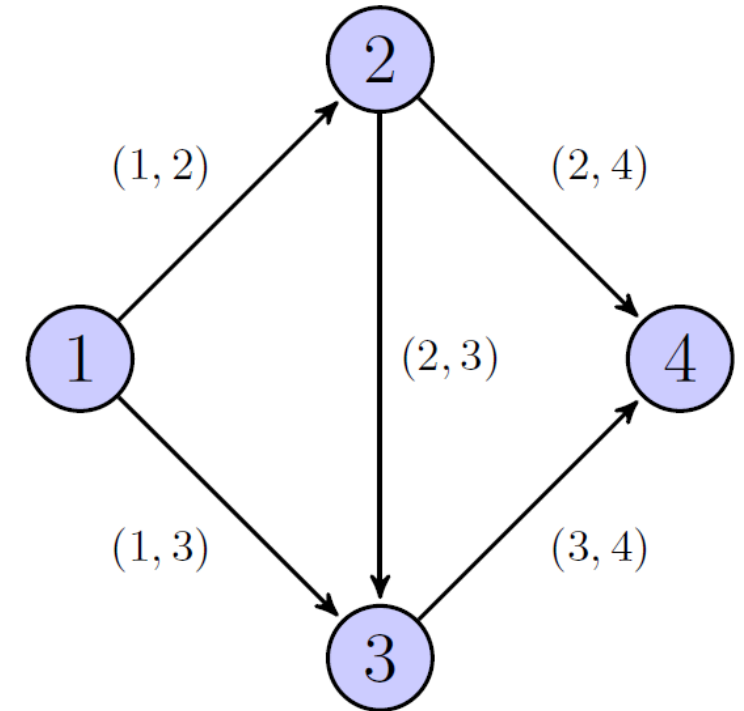


Figure 2.1: Example network notation

# Path

► A **path**  $\pi$  is a sequence of  $k$  adjacent links connecting node  $i_0$  to  $i_k$ .

► denoted either by specifying the links:

$$\pi = \{(i_0, i_1), (i_1, i_2), (i_2, i_3), \dots, (i_{k-1}, i_k)\}$$

► or more compactly by specifying the nodes :

$$\pi = [i_0, i_1, i_2, i_3, \dots, i_{k-1}, i_k]$$

► **Example:** there are three paths between nodes 1 and 4:

►  $\pi_1 = [1, 2, 4]$

►  $\pi_2 = [1, 3, 4]$

►  $\pi_3 = [1, 2, 3, 4]$

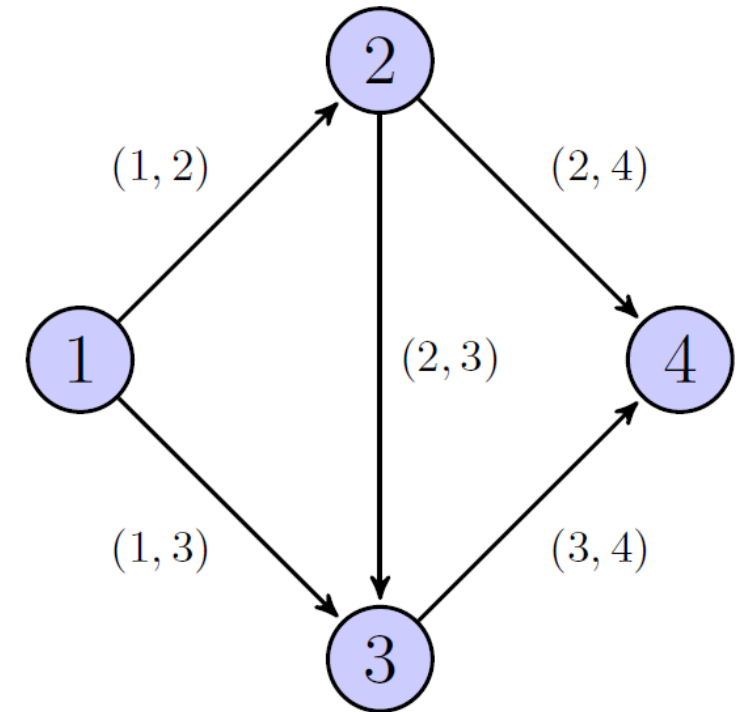
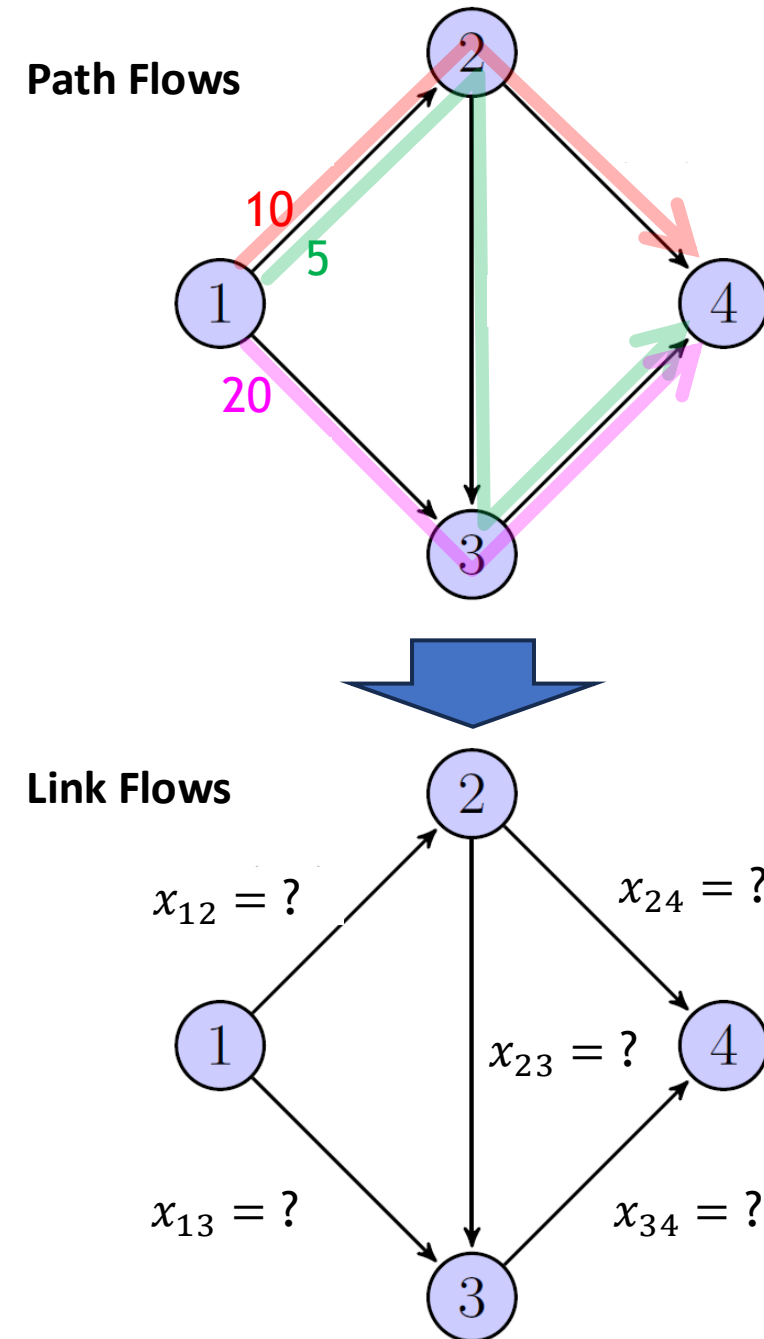


Figure 2.1: Example network notation

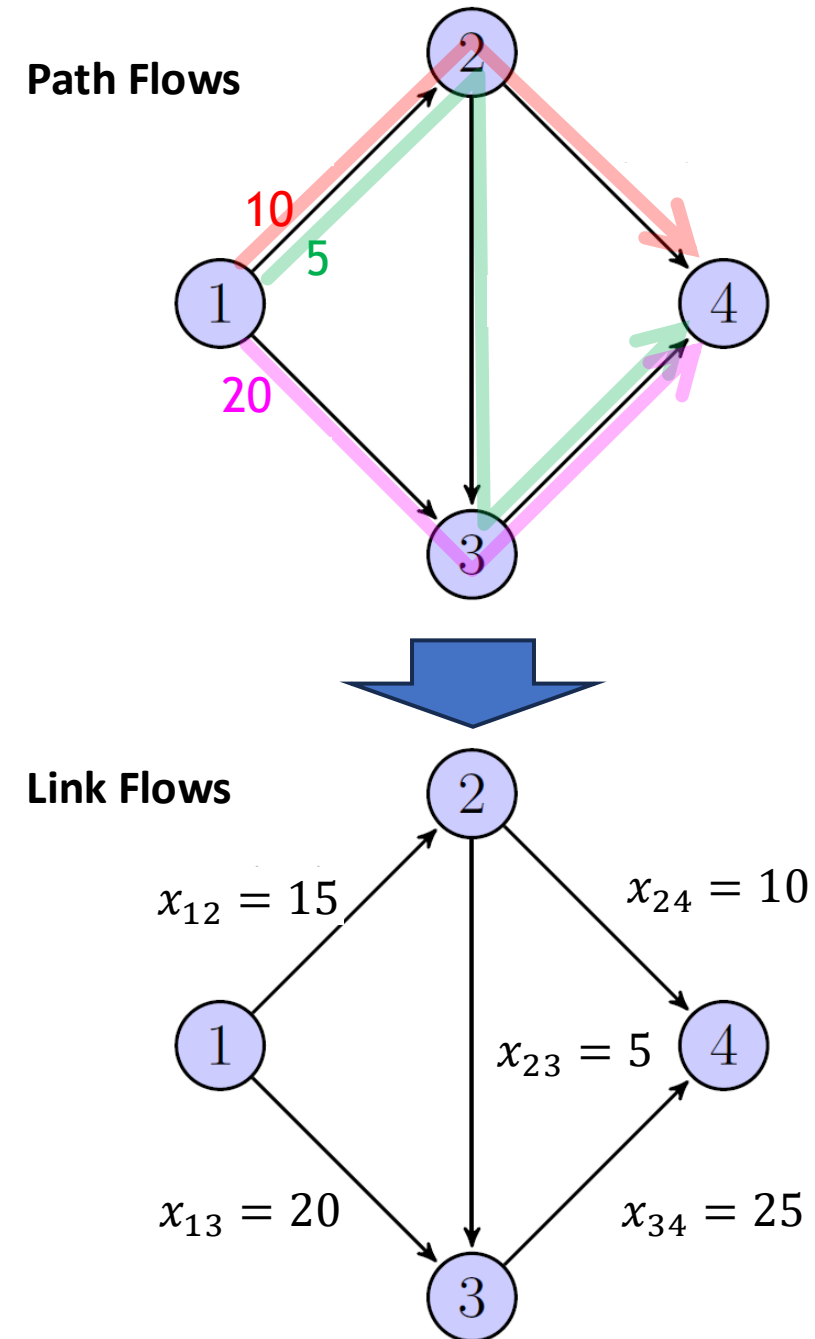
# Path Flow and Link Flow

- ▶ Path flows along the three paths between nodes 1 and 4:
  - $\pi_1 = [1, 2, 4]$ : path flow = 10
  - $\pi_2 = [1, 3, 4]$ : path flow = 20
  - $\pi_3 = [1, 2, 3, 4]$ : path flow = 5
- ▶ The resulting link flows ( $x_{ij}$ ) are:
  - Link (1,2): link flow  $x_{12} = ?$
  - Link (1,3): link flow  $x_{13} = ?$
  - Link (2,3): link flow  $x_{23} = ?$
  - Link (2,4): link flow  $x_{24} = ?$
  - Link (3,4): link flow  $x_{34} = ?$



# Path Flow and Link Flow

- ▶ Path flows along the three paths between nodes 1 and 4:
  - $\pi_1 = [1, 2, 4]$ : path flow = 10
  - $\pi_2 = [1, 3, 4]$ : path flow = 20
  - $\pi_3 = [1, 2, 3, 4]$ : path flow = 5
- ▶ The resulting link flows ( $x_{ij}$ ) are:
  - Link (1,2): link flow  $x_{12} = 10 + 5 = 15$
  - Link (1,3): link flow  $x_{13} = 20$
  - Link (2,3): link flow  $x_{23} = 5$
  - Link (2,4): link flow  $x_{24} = 10$
  - Link (3,4): link flow  $x_{34} = 5 + 20 = 25$

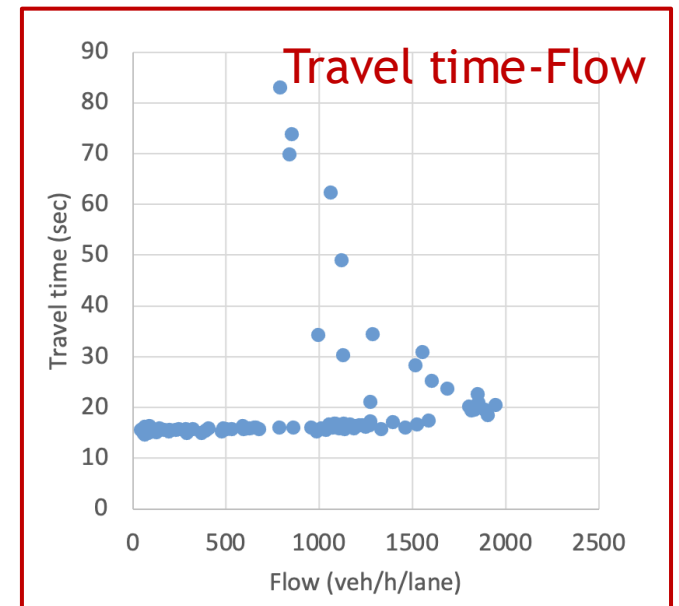
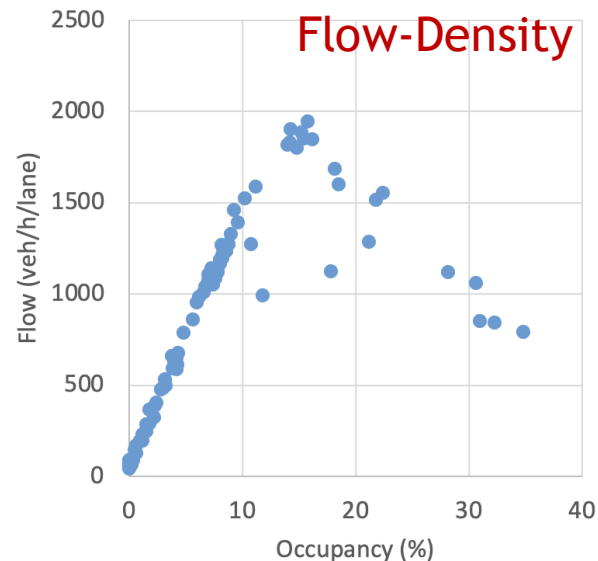
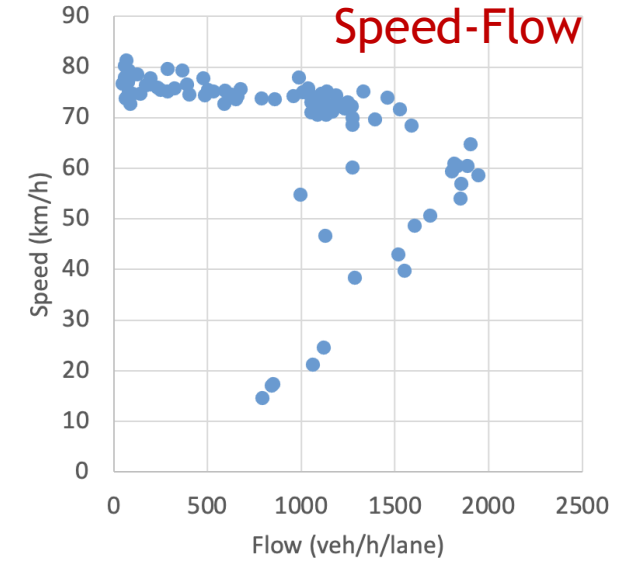
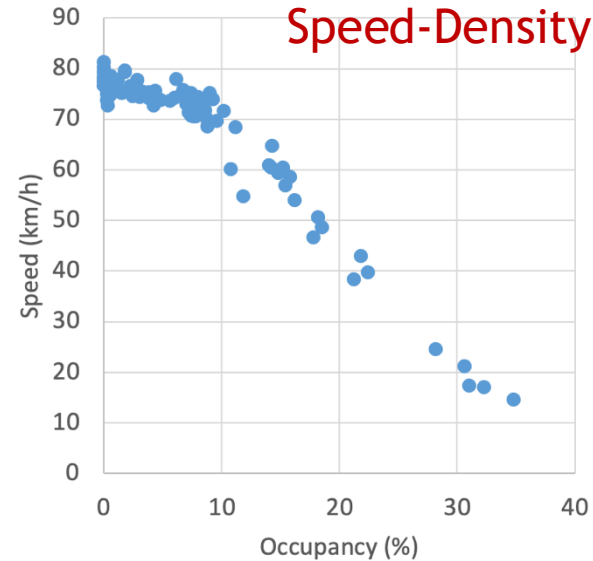


# Network Terminology

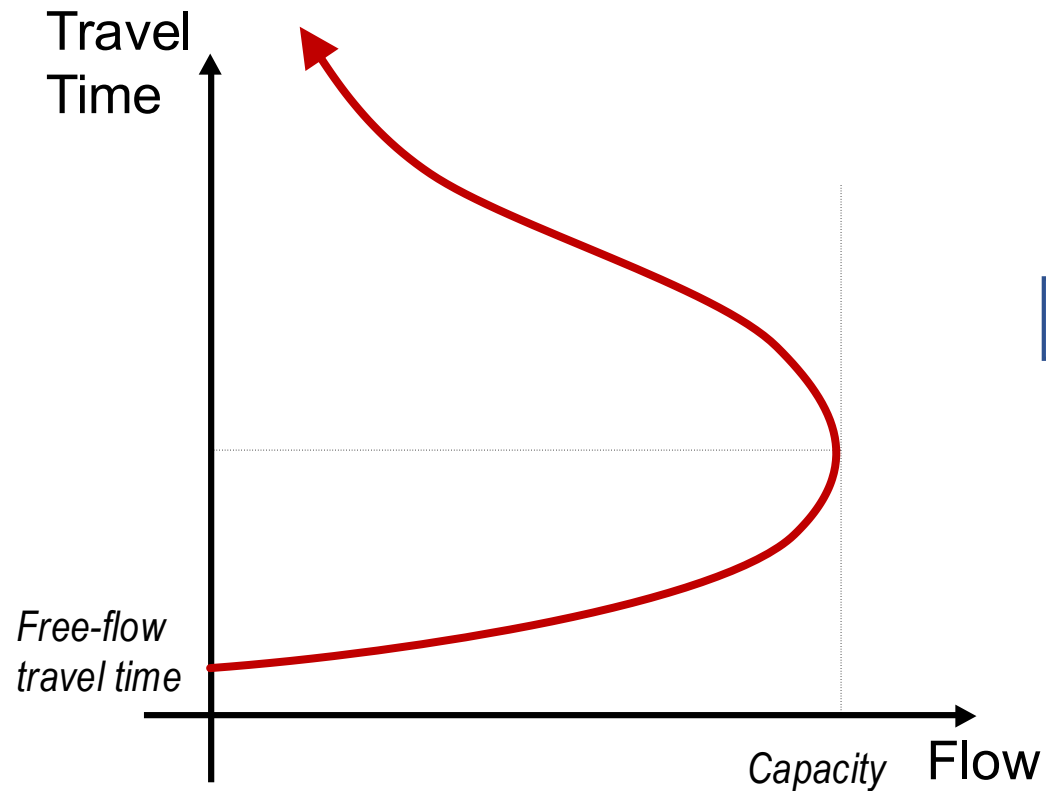
| Terminology in this course | Alternative terms |
|----------------------------|-------------------|
| <b>Network</b>             | Graph             |
| <b>Node</b>                | Vertex            |
| <b>Link</b>                | Arc, Edge, Line   |
| <b>Path</b>                | Route             |
| <b>Flow</b>                | Volume            |

# Link Performance Function

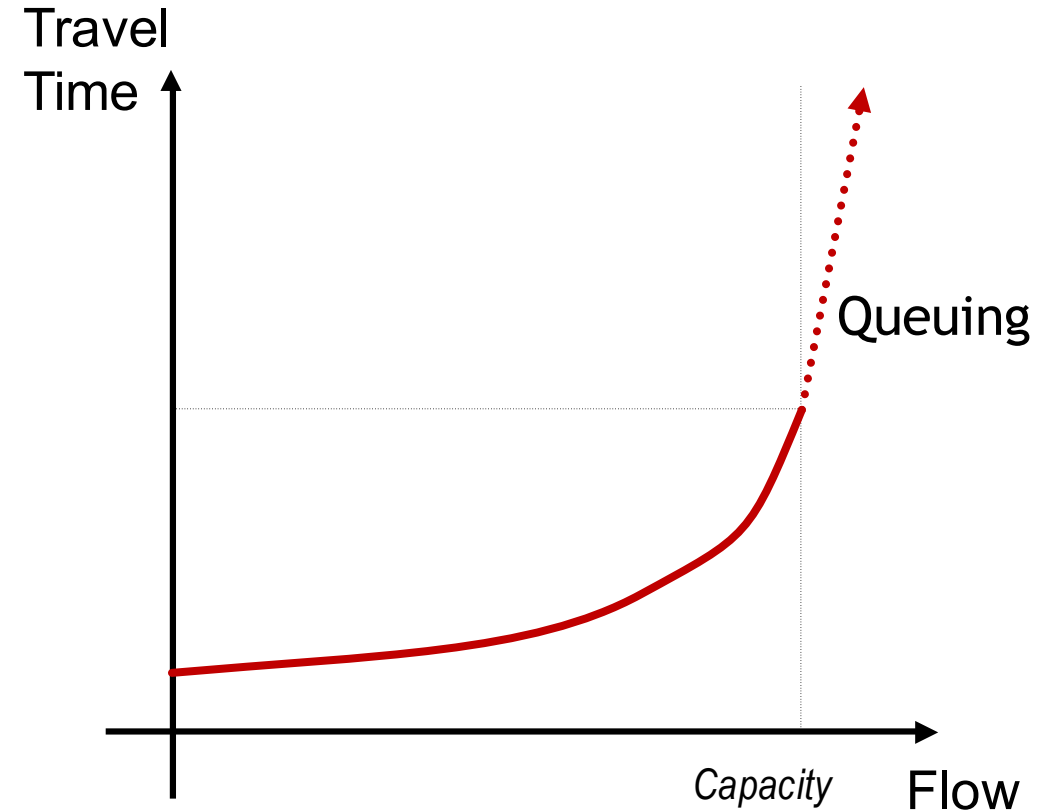
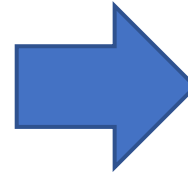
- ▶ *Fundamental Diagram* of link flow, density, and speed.
- ▶ *Speed-flow relationship* leads to the relationship between **travel time** and **flow**.
  - Travel time =  $\frac{\text{Travel distance}}{\text{Speed}}$
- ▶ **Link Performance Function**,  $t(x)$ , describes the level of congestion, i.e., link travel time ( $t$ ) as a function of link flow ( $x$ ).



# Link Performance Function



Cannot be used as a link performance function as there are two travel times at any given flow



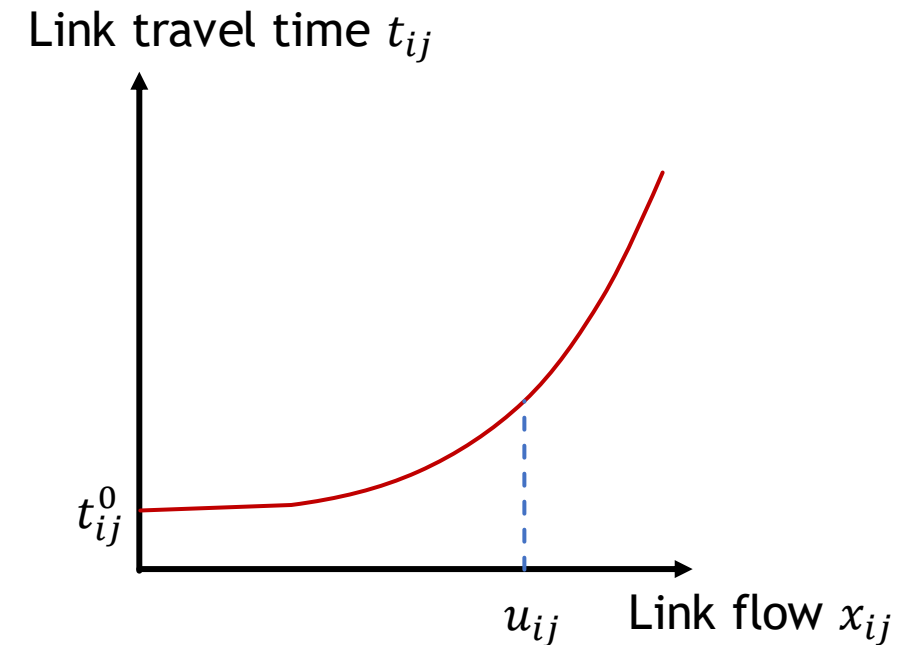
For modelling purposes, a functional form without the backward-bending part has been traditionally adopted, mainly capturing travel time increasing more quickly as traffic flow approaches capacity

# Bureau of Public Roads (BPR) Function

- ▶ The most popular link performance function is the BPR function, originally proposed by engineers at the U.S. Bureau of Public Roads in 1960.

$$t_{ij}(x_{ij}) = t_{ij}^0 \left[ 1 + \alpha \left( \frac{x_{ij}}{u_{ij}} \right)^\beta \right]$$

- $t_{ij}^0$  : the free-flow travel time on link  $(i, j)$
- $u_{ij}$  : the capacity of the link
- $\alpha$  and  $\beta$  : model parameters that can be calibrated to data  
( $\alpha = 0.15$  and  $\beta = 4$  are commonly used)





# Traffic Assignment

- Conceptual framework: The Notion of ***Equilibrium***

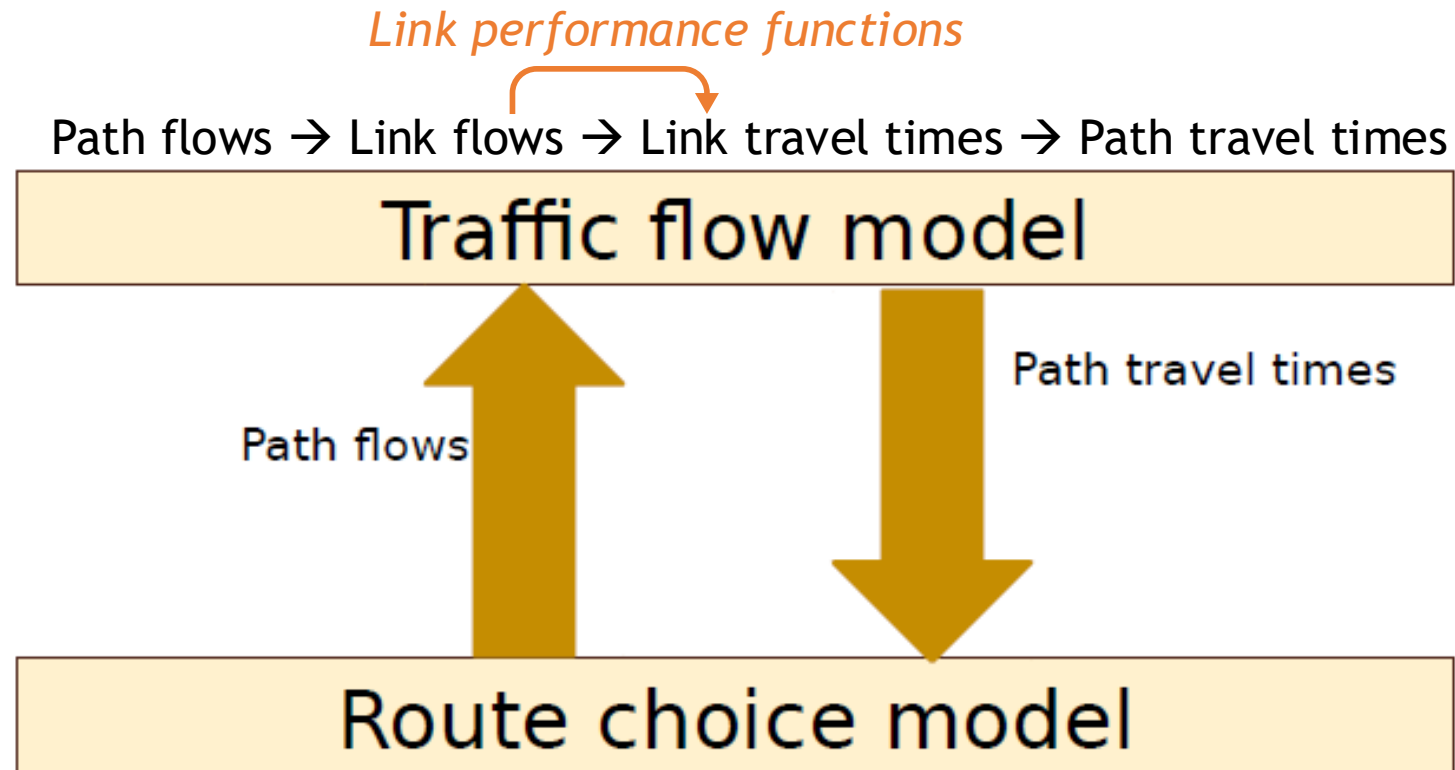
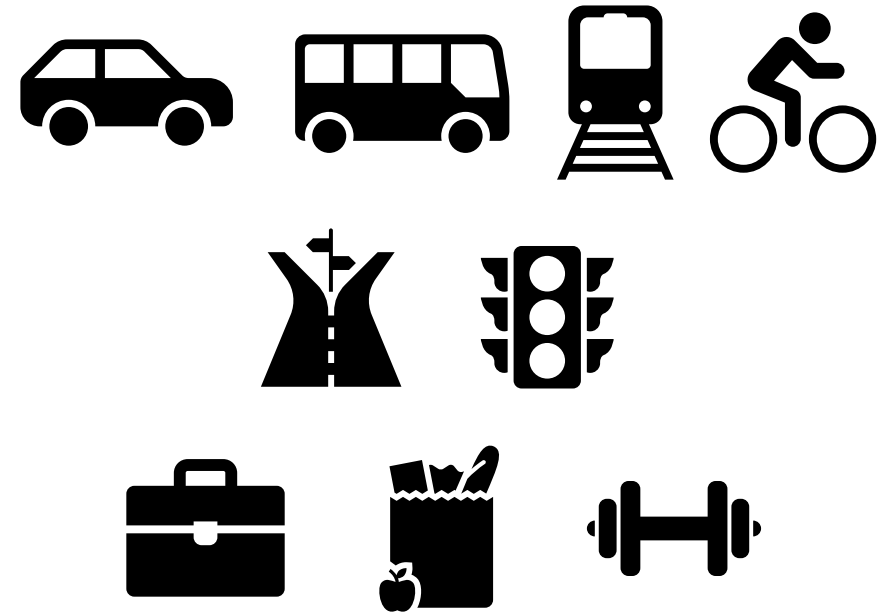


Figure 1.3: Traffic assignment as interacting systems.

# Discrete Choice Modelling

# Travel Choices in Transport Networks

- ▶ Travellers make decisions about:
  - Mode choice (e.g., car, bus, train, cycling)
  - Route choice (e.g., shortest path, motorway vs. arterials, toll vs. free road)
  - Departure time choice (e.g., peak vs. off-peak)
  - Destination choice (e.g., workplace, shopping, gym)
  - Trip generation and activity participation
- ▶ These choices influence overall travel demand, network performance, and congestion patterns.



# Why Model Travel Choices?

- ▶ Travel choices **directly impact** transport network flows.
- ▶ **Traffic assignment models** require demand inputs, which come from travel behaviour models.
- ▶ Understanding choices allows us to:
  - Predict **future demand** under different scenarios (e.g., new transport policies, infrastructure changes)
  - Evaluate the **impact of demand management strategies** (e.g., toll roads, congestion pricing)
  - Design effective **public transport systems**.

# Discrete Choice Modelling

- ▶ A **behavioural framework** to model decision-making among discrete alternatives.
- ▶ Based on **Random Utility Theory** (RUT):
  - Each alternative has an associated utility.
  - The traveller chooses the option with the highest perceived utility.
- ▶ Probabilistic in nature: captures **variability** in individual choices.

# Basic Ideas in Discrete Choice Modelling

## ► Utility Function:

$$U_{ni} = V_{ni} + \epsilon_{ni}$$

where

$U_{ni}$  = true utility that decision maker  $n$  obtains from alternative  $i$

$V_{ni}$  = observed deterministic component of  $U_{ni}$  (function of attributes such as travel time, cost)

$\epsilon_{ni}$  = unobserved random component of  $U_{ni}$

e.g.,

$$V_{car} = \alpha Time_{car} + \beta Cost_{car}$$

$$V_{bus} = \alpha Time_{bus} + \beta Cost_{bus}$$

## ► Multinomial Logit Model (most common form):

### ➤ Probability of choosing alternative $i$ :

$$P_{ni} = \frac{e^{V_{ni}}}{\sum_j e^{V_{nj}}}$$

### ➤ Ensures probabilities sum to 1 and respect independence assumptions.

## ► Extensions: Nested Logit, Mixed Logit, etc.

# What is Involved in Discrete Choice Modelling?

- ▶ The process of building a discrete choice model involves:
  - **Choice data collection**
    - Observed or surveyed choice data, data on alternative choices, decision-maker characteristics
  - **Model specification**
    - Model type, utility function specification
  - **Parameter estimation**
    - Estimation of the parameters in utility functions via Maximum Likelihood Estimation (MLE)
  - **Model validation & Interpretation**
    - Sign and magnitude of parameters, elasticities, value of time (VOT)
  - **Application in transport demand modelling**
    - Demand forecasting, traffic assignment integration, evaluation of impacts of policies, new routes/modes, etc.

Thank you